Transfer Entropy on a Microprocessor
Supplementary appendix

Priyongkar Kundu, Honours thesis, University of Sydney, 2026

This supplementary appendix reproduces the thesis's full appendix as it stood before the detail tables were moved out of the thesis to shorten it. Tables and figures are numbered S1, S2, ...; the thesis cites the moved items by these numbers. References to chapters, sections and equations point to the thesis.

CHAPTER S

Supporting tables and data sources

Every table and figure in this appendix is cited from the main text. Tables marked † were computed after the final run from its saved outputs (Section 4.1.2). Table S1 lists the data file behind every table and figure of the thesis.

S.1 Data sources

Table S1: Data sources of the main-text tables and figures, and the state of each source. Paths are relative to the project's data directory; the code and outputs are also in the code repository (Section 4.1.2), whose `docs/file_renames.md` maps these paths to its own. `RUN`: `results/v5_bitcorrect_every_third_mi_ais_te_k1_12_L1_d1_ais_x1/thesi` `NT`: `analyses/new_tests/output/`. `RP`: `analyses/replication/output/`. `SR`: `analyses/supplementary_recompute/output/`. State: *run*, written by the main run; *logged*, written by a hash-logged test; *recomputed*, recomputed from the run's saved outputs after the original file was lost (Appendix S.12); *not retained*, the value first reported by a supplementary script whose output file was not kept.

Item	Source	State
Schematic figures (1.1, 2.1, 2.2)	No data	–
Tab. 4.1, 4.2	Run notebook `notebooks/Honours_v5_executed.ipynb`; design files `analyses/new_tests/PREDICTIONS*.md`, `analyses/replication/PREDICTIONS_*.md`	run; logged
Regeneration, change coding, truncation (Sec. 4.3.1)	`analyses/pitfall/CONVERSION_NOTE.md`; `data/regen/*.regen.ts.spikes.npy` and `.json`	logged
Tab. 5.1: lag-1 TE, lag-0 CMI, MI	`RUN/tables/rq1_discrimination.csv`; within-target and parity rows `SR/recomputed_supplementary.json` (`P_discrimination_supplementary`)	run; recomputed
H1 robustness (Sec. 5.2)	`SR/h1_robustness_checks.json`	recomputed

Item	Source	State
Fig. 5.2, Tab. S27	`SR/recomputed_supplementary.json` (`P_strata`)	recomputed
Tab. 5.2, S38, Sec. 5.3.3	`NT/recording_results.json` (A3), `NT/A9_results.json`, `NT/A12_results.json` (A12, A14, A18), `NT/A13_A16_chip_results.json`	logged
Tab. 5.3	`RP/R2_Pitfall_results.json`, `RP/R2_SpaceInvaders_results.json`	logged
Fig. 5.3	`NT/A9_results.json`, `NT/A11_results.json`, `NT/A12_results.json`, `RP/R2_*_results.json`; script `analyses/figures_v11/fig_followership_and_locking.py`	logged
Tab. S33	`NT/A11_results.json`, `NT/A15_results.json`	logged
Tab. S36, S37, Fig. 5.5	`NT/chip_results.json`, `NT/A7_results.json`, `NT/supp_harder.json`, `NT/A13_A16_chip_results.json`; open/closed shares `NT/supp_a16_open_closed.log`	logged
Tab. 5.4, S35, A20 (Sec. 5.6.2)	`NT/recording_results.json`, `NT/supp_latch_vs_passnb.json`, `NT/A12_results.json` (A13, A19); `RP/A20_results.json`	logged
Tab. S29	`NT/nmos_results.json`, `NT/A10_results.json`	logged
Tab. 5.5, S34; remedy (Sec. 5.7.2)	`SR/recomputed_supplementary.json` (`B_baselines_*`, `B_within_*`, `P_*`); phase class only and the physical-fitted covariate model from `SR/recomputed_supplementary_extras.json`, which supersedes the first pass's definitions	recomputed
Tab. S30, S31, S22, S23; Tab. S50, rows H1–H4	`RUN/tables/rq1_direction.csv`, `rq1_signed_effect.csv`, `rq1_paired_contrasts.csv`; `SR/recomputed_supplementary_pass3.json`	run; recomputed
Tab. S28	`RUN/tables/rq1_te_by_relation.csv`	run
Tab. 6.1	`RUN/tables/rq2_network_comparison.csv`, `rq2_paired_differences.csv`	run
Fig. 6.1, Tab. S47, S43, S44	`SR/recomputed_supplementary.json` (`D_*`, `E_*`), `SR/recomputed_supplementary_extras.json` (`holm_joint`)	recomputed
Tab. S45, S46	`RUN/tables/rq2_target_summaries.csv`; pass-neighbour traces	run; not retained
Tab. 6.2; A4, R1	`NT/idtxl_results.json`	logged

Item	Source	State
Tab. 6.3, S41	`SR/h6_run_of_record_R1.json`, `SR/h6_run_of_record_R1_details.json` (from `NT/R1_targets/`); A17 `NT/A17_results.json`	recomputed; logged
Validation tables and figures (Chapter 4)	`RUN/tables/null_calibration_by_target.csv`, `synthetic_summary.csv`, `sim_representation_scores.csv`, `sim_idtxl.csv`, `nmos_summary.csv`; calibration re-weighting and placebo rows, chip-simulation intervals, corrected pairwise-BH rows	run; not retained
Tab. 1.1, S39, 7.1, 7.2, 7.3	As for the tables their rows cite	–

TABLE S2: †Check of the column-to-netlist mapping by gate-node co-grouping, and of the time base by the transistors gated by the clock nodes (Sections 4.3 and 4.3.1).

Check	Value
gate-node sharing: active transistor pairs sharing a gate node	62,387
gate-node sharing: fraction in the same activity group	1.0
gate-node sharing: gate nodes with at least two active transistors	453
gate-node sharing: such nodes whose transistors fall in more than one group	0
random active transistor pairs: fraction in the same group	0.028
transistors gated by cp1	105
cp1-gated transistors: event rate per reduced sample (min, median, max)	0.9895, 0.9895, 0.9895
cp1-gated transistors: groups	one (group 6)
transistors gated by cclk	273
cclk-gated transistors: event rate per reduced sample (min, median, max)	0.9895, 0.9895, 0.9895
cclk-gated transistors: groups	one (group 6)
transistors gated by clk0	2
clk0-gated transistors: event rate per reduced sample (min, median, max)	0.9895, 0.9895, 0.9895
clk0-gated transistors: groups	one (group 6)

TABLE S3: †Test of the event semantics of the recording. Under off-to-on events an inverter's input transistor and the transistors its output gates switch on at different times; under any-change events they change together (Section 4.3.1).

Quantity	Value
Nodes driven by a single pull-down transistor and no pass transistor (inverters), outside the clock group	326
Driver-reader transistor pairs at those nodes, both active	1,201
of which with identical event series (same activity group)	1,201 (100%)
of which outside the clock group	1,166
of which with events in fewer than 2% of samples	285, all identical
Example: Y register bit y1 and its complement noty1	identical series, events in 0.42% of samples
Nodes with one pull-down and at least one pass transistor: driver-reader pairs identical	422 of 473
Nodes with several pull-downs (NOR gates): driver-reader pairs identical	2,111 of 3,989

S.2 Recording, grouping and reference

TABLE S4: †Time-base and stationarity checks of the recording (Sections 4.3.1 and 4.3.2).

Check	Value
Spearman correlation of group event rates, samples 5,000-64,999 vs 65,000-133,925	1.000
Fraction of groups whose rate differs more than twofold between the two halves	0.004
Spearman correlation of group rates, first 5,000 samples vs remainder	0.873
Median coefficient of variation of group rate across 5,000-sample windows (windows 2-27)	0.251
Fraction of groups with window-to-window CV above 0.5	0.297
Phase-0 share of events, first 30,000 original rows	0.861
Phase-0 share of events, minimum over later 30,000-row windows	0.999
Aggregate row-activity autocorrelation at 3 original rows	0.921
Aggregate row-activity autocorrelation at 6 original rows	0.978
Aggregate row-activity autocorrelation at 9 original rows	0.915
Aggregate row-activity autocorrelation at 12 original rows	0.972
Within-column inter-event gaps of exactly 3 original rows	69,911,954
Within-column inter-event gaps of exactly 6 original rows	12,168,175
Within-column inter-event gaps of exactly 9 original rows	1,461,562
Within-column inter-event gaps of exactly 12 original rows	6,381,073
Within-column inter-event gaps not a multiple of 3 (fraction)	1.16e-04

TABLE S5: †Alignment of the every-third-row reduction with the clock (Section 4.3.2). Upper panel: raw-recording events of the clock node `clk0` by row phase (row index modulo 3; the reduction keeps phase 0). Lower panel: AUC of the three pairwise scores (95% target-cluster intervals) under four treatments of the opening samples; AIS histories reselected in each. The clock group's reduced event rate is 0.72 in samples 0–4,999 and 1.00 afterwards.

Row phase	Raw rows	`clk0` events
0	0–0	1
2	2–2	1
1	4–514	171
0	516–516	1
2	518–4,220	1,235
1	4,222–4,222	1
0	4,224–401,775	132,518

Score	Treatment of the opening	Physical AUC	Group AUC	Within-target AUC
lag-1 TE	Whole recording, original reduction	0.500 [0.411, 0.596]	0.469 [0.445, 0.493]	0.434 [0.418, 0.450]
	Samples 1–1,407 re-reduced in step with the clock	0.438 [0.351, 0.528]	0.395 [0.371, 0.420]	0.316 [0.295, 0.338]
	Clock-aligned (samples 1,408 onwards)	0.440 [0.352, 0.530]	0.396 [0.372, 0.421]	0.320 [0.299, 0.342]
	Samples 5,000 onwards	0.439 [0.352, 0.529]	0.394 [0.370, 0.420]	0.320 [0.299, 0.341]
lag-0 CMI	Whole recording, original reduction	0.593 [0.504, 0.683]	0.632 [0.611, 0.651]	0.632 [0.613, 0.651]
	Samples 1–1,407 re-reduced in step with the clock	0.526 [0.440, 0.613]	0.560 [0.537, 0.582]	0.509 [0.477, 0.540]
	Clock-aligned (samples 1,408 onwards)	0.527 [0.441, 0.614]	0.561 [0.538, 0.582]	0.511 [0.479, 0.541]
	Samples 5,000 onwards	0.527 [0.442, 0.614]	0.560 [0.536, 0.581]	0.511 [0.479, 0.542]
zero-lag MI	Whole recording, original reduction	0.545 [0.498, 0.591]	0.597 [0.578, 0.615]	0.615 [0.594, 0.636]
	Samples 1–1,407 re-reduced in step with the clock	0.501 [0.415, 0.590]	0.516 [0.495, 0.537]	0.490 [0.461, 0.521]
	Clock-aligned (samples 1,408 onwards)	0.501 [0.416, 0.590]	0.516 [0.495, 0.537]	0.492 [0.462, 0.522]
	Samples 5,000 onwards	0.502 [0.416, 0.590]	0.516 [0.494, 0.536]	0.492 [0.463, 0.523]

TABLE S6: †Structure of the activity groups and of the netlist edges that lie within them (Sections 4.3.3 and 4.4).

Quantity	Value
Within-group edges	4,815
of which pull-down	3,734
of which pass	1,081
Groups containing within-group edges	240
Within-group edges inside the largest group	2,182
Largest group: transistors	456
Largest group: distinct gate nodes	22
Largest group: samples with an event (of 133,926)	132,520
Median event count of the groups hosting within-group edges (edge-weighted)	35,830
Multi-member groups	347
Multi-member groups whose members all share one gate node	78
Singleton groups	172
Median group event rate (fraction of samples)	0.073
Median transistor event rate (active transistors)	0.104

S.3 Supporting material for Chapter 2

Proposition 1 (combinational wiring and lag-1 transfer entropy). *Let units be recorded as node states, and let each sample be long enough for combinational logic to settle within it. Let Y be the output of a logic gate whose inputs are X and W (W may collect several nodes), connected by combinational logic with no latch in between, so that $Y_t = g(X_t, W_t)$ for a Boolean function g. Then, for any target history $Y^- = Y_t^{(k)}$,*

$$T_{X\to Y}(k,1,1) = I\big(X_{t-1};\, Y_t \mid Y^-\big) \le I\big(X_{t-1};\, X_t, W_t \mid Y^-\big). \tag{S.1}$$

PROOF. Because Y_t is a function of (X_t, W_t),

$$I(X_{t-1}; Y_t \mid X_t, W_t, Y^-) = 0.$$

Expanding $I(X_{t-1}; X_t, W_t, Y_t \mid Y^-)$ by the chain rule in the two orders, and using this zero, gives

$$I\big(X_{t-1};\, Y_t \mid Y^-\big) = I\big(X_{t-1};\, X_t, W_t \mid Y^-\big) - I\big(X_{t-1};\, X_t, W_t \mid Y_t, Y^-\big), \tag{S.2}$$

and the last term is non-negative. □

A source's previous sample can inform a logic gate's output only through what it says about the gate's current inputs; Equation (S.2) gives the slack. Three points fix the scope. First, the bound is not specific to wiring: an unwired source, with $Y_t = h(W_t)$, obeys the same inequality, since

$$I\big(X_{t-1};\, X_t, W_t \mid Y^-\big) = I\big(X_{t-1};\, W_t \mid Y^-\big) + I\big(X_{t-1};\, X_t \mid W_t, Y^-\big), \tag{S.3}$$

so the inequality orders nothing by itself. Second, combinational ancestors of Y within the phase have one route to lag-1 dependence that other sources lack. The decomposition

$$I\big(X_{t-1};\, X_t, W_t \mid Y^-\big) = I\big(X_{t-1};\, X_t \mid Y^-\big) + I\big(X_{t-1};\, W_t \mid X_t, Y^-\big) \tag{S.4}$$

holds for any source; its first term is the source's memory, and its second can be positive for an ancestor through a collider, since Y^- contains $Y_{t-1} = g(X_{t-1}, W_{t-1})$, a common effect of the source's previous sample and the other inputs. For the running NOR gate, $A_{t-1} = 0$ with $Q_{t-1} = 0$ reveals $B_{t-1} = 1$. Third, the proposition concerns node states and lag 1 only; for change indicators, ΔY_t has X_{t-1} as an argument, so on the change-coded recording the relevant result is Proposition 2.

Heuristic consequence. For combinational wiring recorded as states, lag-1 transfer entropy can register the connection only through the two terms of Equation (S.4), whereas lag-0 CMI is not subject to the bound, because X_t is an argument of g. If within-phase dependence is dominated by direct coupling rather than co-drive, an assumption the proposition does not supply, lag-0 dependence should identify combinational wiring better than lag-1 transfer entropy. That is the prediction of H2; the proposition motivated it before the run and decides nothing by itself. Where a clocked pass transistor separates source and target, the proposition does not apply: recorded as states the latch is visible at lag 0 and lag 1 alike, and recorded as changes it is the opposite-phase case that Proposition 2 leaves unconstrained.

Remark 2 (approximate locking). Let $E_t = 1$ when $\Delta X_{t-u} = 1$ in a sample of parity other than p_X or $\Delta Y_t = 1$ in a sample of parity other than p_Y, and let $\varepsilon = \Pr(E_t = 1)$. By the chain rule,

$$I(\Delta X_{t-u}; \Delta Y_t \mid P_t, Z_t) \le I(\Delta X_{t-u}; E_t \mid P_t, Z_t) + I(\Delta X_{t-u}; \Delta Y_t \mid E_t, P_t, Z_t).$$

The first term is at most $h(\varepsilon)$; in the second, the case argument of the proof of Proposition 2 applies whenever $E_t = 0$, so only samples with $E_t = 1$ contribute, at most ε times one bit. Hence

$$I\big(\Delta X_{t-u};\, \Delta Y_t \,\big|\, Z_t\big) \le I\big(\Delta X_{t-u};\, P_t \,\big|\, Z_t\big) + h(\varepsilon) + \varepsilon.$$

For a pair whose source and target each have 5 per cent of their events off phase, ε is at most $0.05(r_X + r_Y)$, with r the event rate per sample. At $\varepsilon = 0.001$, $h(\varepsilon)+\varepsilon$ is about 12 millibits, an order of magnitude

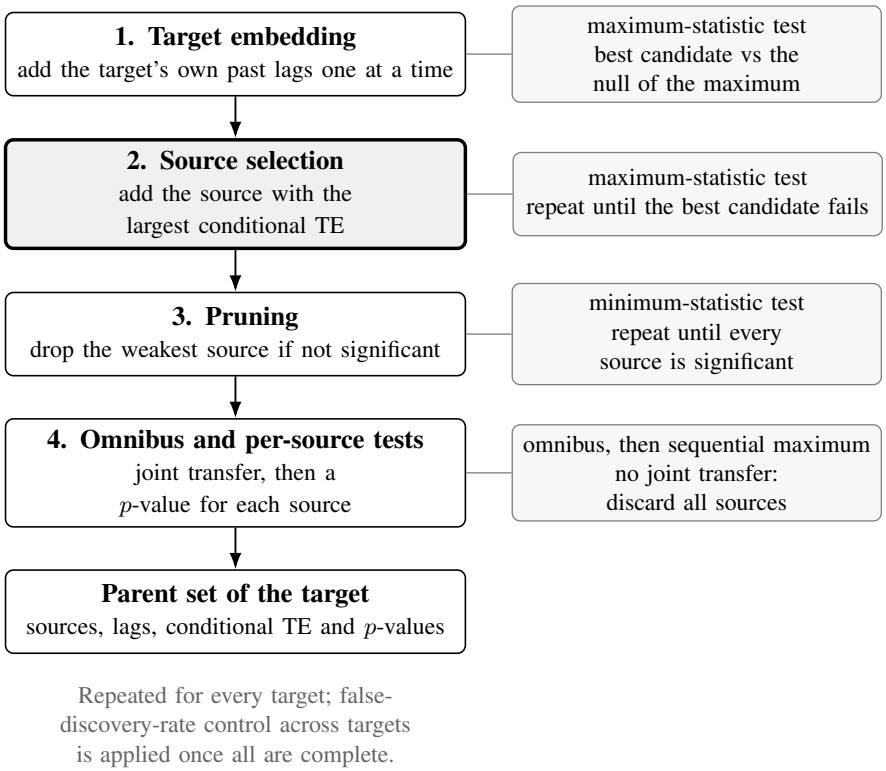


FIGURE S1: Greedy multivariate transfer-entropy inference for one target (Lizier and Rubinov, 2012; Novelli et al., 2019), as implemented in IDTxl. The statistical tests on the right distinguish it from a plain greedy search. The emphasised stage, source selection, is where conditioning on already-selected sources acts and where Chapter 6 locates the multivariate gain.

above the mean lag-1 transfer entropy of same-phase group pairs, so the bound is uninformative for approximately locked pairs, and Section 5.3.1 measures how often they exceed $B_{X\to Y}$ instead.

The periodic-drive prediction. Before the calibration of Section 4.6, the thesis predicted from an effective-sample-size argument that shared periodic drive would make the analytic test reject too often, most for the most periodic units. Section 4.6 tests that prediction, and Section 4.6 sets out why the argument was mis-specified: the autocorrelation of a strictly phase-locked change indicator does not decay, so the effective-sample-size sum does not converge, and the excess rejections the calibration observes trace to dependence on the phase that the target's history does not absorb.

S.3.1 Supporting material for Chapter 4

This section holds the detail behind Chapter 4: the analysis pipeline, the structural reference and its relation classes and the statistics of H3 and H4, one further check of the reduction, and the detail of the three parts of the validation (Section 4.6). The body states the conclusion of each item.

S.3.1.1 The analysis pipeline

Figure S2 draws the path from the recording and the netlist to the evaluation that Chapter 4 follows.

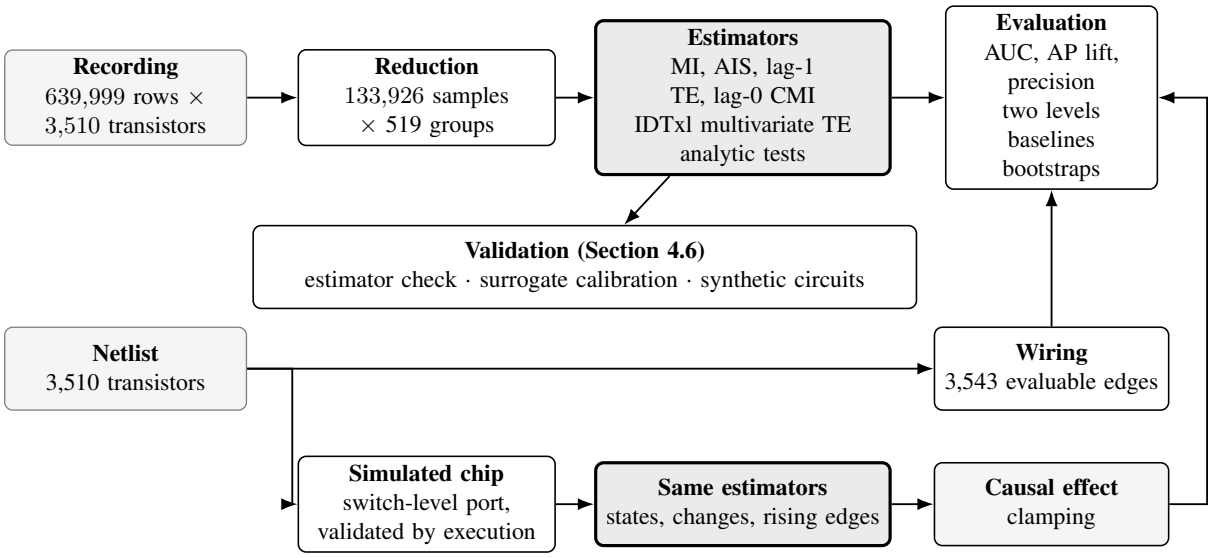


FIGURE S2: The analysis pipeline: the recording is reduced and grouped, scored by the information measures and evaluated against the wiring, while the netlist also drives the simulated chip, which adds an interventional reference. The recording is reduced to one sample per clock phase and merged into activity groups. Boxes with heavy outlines mark where information measures are computed; lightly shaded boxes are data sources and the interventional reference. MI, mutual information; AIS, active information storage; TE, transfer entropy; CMI, conditional mutual information; AUC, area under the receiver operating characteristic curve; AP, average precision.

TABLE S7: Structural relation classes of the 268,842 ordered group pairs. Each pair takes the first class that applies, in the order listed. †Supplementary.

	Transistor-path definition (pre-specified)		Group-graph definition†	
Class	Condition on some transistor pair	Group pairs	Condition on the group graph	Group pairs
Wired	Edge from a transistor of A to one of B	1,437	Edge $A \to B$	1,437
Reverse only	Edge from B to A only	1,437	Edge $B \to A$ only	1,437
Two-step	Path of two edges from A to B	690	$A \to C \to B$ for a group C	1,364
Shared parent	A transistor with edges to both	227	A group C with $C \to A$ and $C \to B$	76,546
Shared child	A transistor with edges from both	1,155	A group C with $A \to C$ and $B \to C$	439
Unrelated	None of the above	263,896	None of the above	187,619

S.3.1.2 The structural reference

Figure S3 draws the two edge types of Section 4.4, the pass-through relation of Section 4.4, and the split of the netlist's edges behind the observability ceiling (Section 4.4).

S.3.1.3 Structural relation classes

Table S7 gives the counts behind the relation classes of Section 4.4.

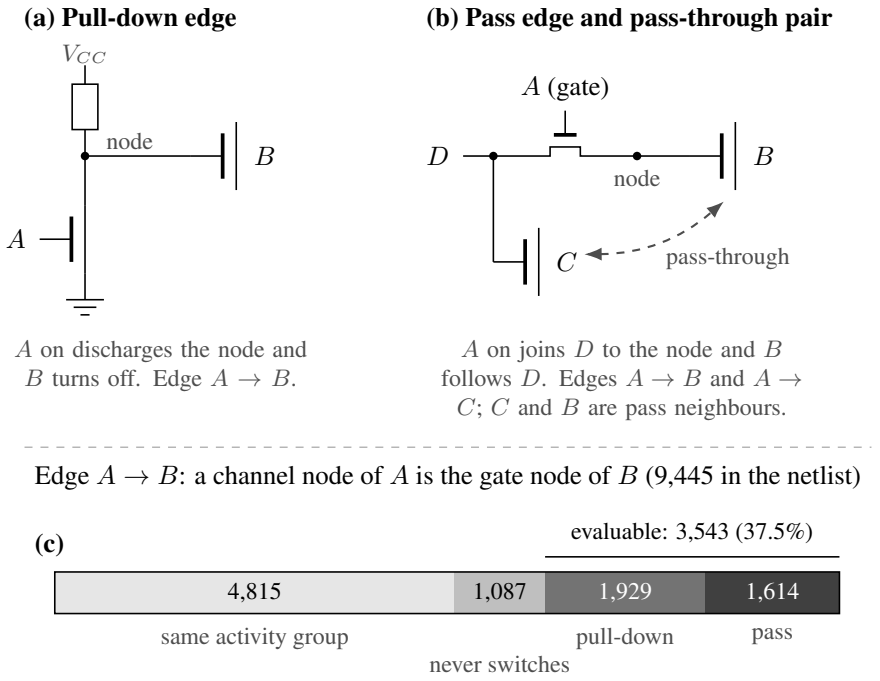


FIGURE S3: The structural reference: an edge runs from a transistor to each transistor whose gate node its channel node drives, and only 3,543 of the 9,445 edges (37.5 per cent) can be evaluated from the activity. (a) A pull-down edge: transistor A discharges the node that gates B. (b) A pass edge: A joins node D to the node that gates B. The reference records edges from the switch A to B and to C, which D gates; the pass-through relation (dashed, defined after the multivariate errors were seen) joins C and B, the images of the two channel nodes of A. (c) The 9,445 edges of the netlist, split into those that cannot be evaluated and the 3,543 evaluable edges by class.

S.3.1.4 The statistics of H3 and H4

Orientation accuracy (H3) is the fraction of the untied wired group pairs for which the orientation with the larger score is the wired one. The pre-specified nuisance predictors were the event rate and the conditional entropy of each member given its own past, each predicting that the member with the larger value is the target; the rate predictor was specified with the wrong sign, and the comparator for H3, chosen after the run as the most accurate of the three once reversed, is the rule that the member with the lower event rate is the target. The *signed influence* (H4) of source X on target Y is

$$\delta = \sum_{s \in \mathcal{S}} w_s \big[p(y_t = 1 \mid s, x_{t-1} = 1) - p(y_t = 1 \mid s, x_{t-1} = 0) \big], \tag{S.5}$$

where $\mathcal{S}$ is the set of the target's history states in which both source values occur and w_s the share of those observations in state s. Under change coding and phase-locked units the wired interaction acts within the phase, so δ measures the sign of the lag-1 association and not that of the wired interaction.

S.3.1.5 Alignment of the reduction with the clock

The supplementary check behind Section 4.3.2 found that reduced samples 1 to 1,407 miss almost every clock change, carrying 0.03 events per sample against 51.8 when reduced in step with the clock, and that the misalignment accounts for their whole difference from the rest of the recording: re-reducing only those samples in step with the clock moves lag-0 CMI's mean within-target AUC from 0.632 to

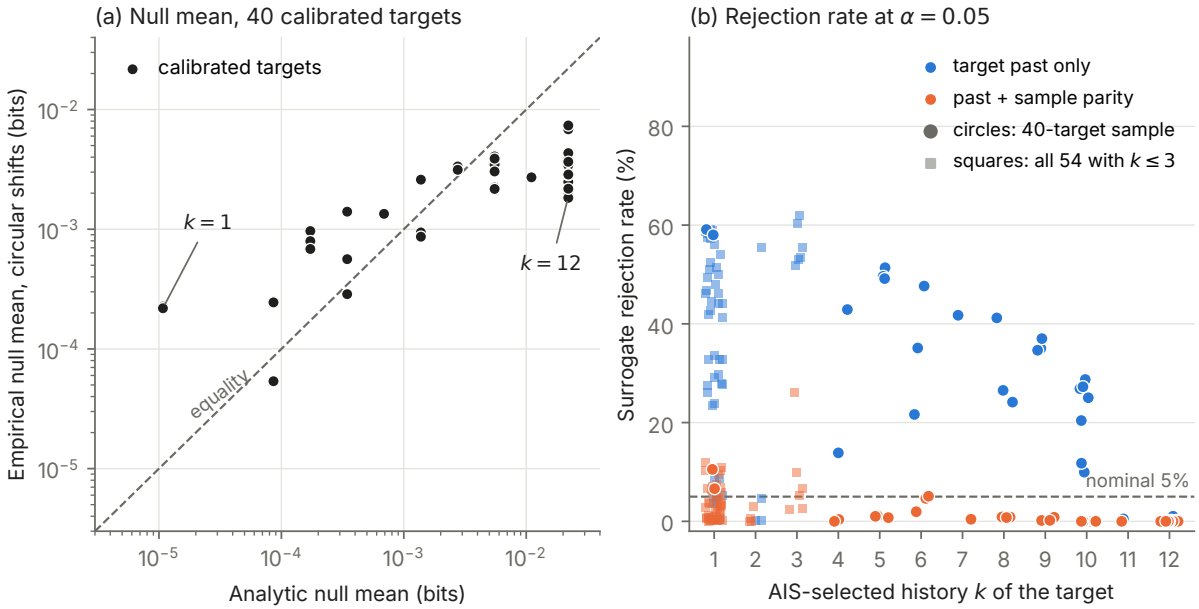


FIGURE S4: The analytic transfer-entropy test rejects 20.5 per cent of circular-shift surrogates on the recording at a nominal 5 per cent, most at short histories, and adding sample parity to the conditioning set lowers the rate at $k = 1$ from 58.6 to 8.6 per cent. (a) Each calibrated target's empirical null mean (100 shifts $\times$ 518 sources) against the analytic null mean, on logarithmic axes; points above the dashed line have an empirical mean larger than the analytic one. (b) Each target's surrogate rejection rate at $\alpha = 0.05$ against its selected history, with the target's past only (blue) and with sample parity added to the conditioning set (orange); circles are the 40-target sample of the main run (exploratory), squares the supplementary[†] calibration of all 54 targets with $k \leq 3$. A calibrated test would place points on the dashed 5 per cent line. Point estimates only; no intervals are drawn.

0.509, close to the 0.511 obtained from samples 1,408 onwards, and dropping further samples changes no AUC by more than 0.005. That about 1 per cent of the samples, nearly free of events, can move a within-target AUC by 0.12 shows how sensitive plug-in scores on sparse binary data are to a block of atypical samples.

S.3.1.6 The calibration of the analytic null in detail

Figure S4 shows the calibration of Section 4.6 target by target. The 40 targets of the main run were drawn with seed 2026 from the 442 targets whose event rate lies between 0.1 and 99.9 per cent, and a supplementary calibration repeated the design for all 54 targets with $k \leq 3$. The empirical null mean is about 21 times the analytic one at $k = 1$ to 3 and a twelfth to a third of it at $k \geq 11$, the only targets below the nominal level. Because 159 targets sit at the cap of $k = 12$, the long-history regime governs nearly a third of the network. At $k = 12$ every calibrated target rejects at most 1.0 per cent of surrogates, yet Bonferroni correction declares 23.9 per cent of real pairs into these targets, dependence that requires the actual alignment; at $k = 1$ about four-fifths of the declarations are matched by phase dependence alone. In both regimes the declared pairs are dependent but mostly not wired.

S.3.1.7 The three supplementary tests of the calibration

On one reading the excess rejections are errors of the chi-squared approximation, from sparse cells at short histories and unvisited states at long ones. On the other, most rejections are correct because the surrogate leaves dependence in place: a circular shift carries a period-two component to the same phase by an even offset and to the opposite phase by an odd one, so a phase-locked target whose short history does not reveal its own phase is predictable from any phase-locked source after any shift. The saved draws cannot separate the readings, because both predict a high rate at short histories and a low one at long ones. Three supplementary analyses do so on all 4,000 surrogate draws (Table S16). Sample parity was added to the target's conditioning set, which should remove phase dependence and leave approximation errors; because doubling the degrees of freedom can itself make the test more conservative, the test was repeated with a placebo in place of parity, a fixed series of independent fair bits that doubles the degrees of freedom identically but carries no phase; and the rejections were split by the parity classes of source and target.

At matched degrees of freedom, conditioning on parity instead of the placebo lowers the sample's mean rejection rate from 13.3 to 0.91 per cent, sixfold or more in every band where the placebo leaves any rejections. For the targets with $k = 1$, where the placebo changes little, parity lowers the rate from 58.6 to 8.6 per cent, and over all 54 targets with $k \leq 3$ from 37.0 to 3.85 per cent. The source-side split agrees: for phase-locked targets, phase-locked sources are rejected more often than mixed-parity sources (23.3 against 11.8 per cent), and with parity conditioned the ordering reverses. One caveat applies at longer histories: where a target's history already reveals its phase, parity adds conditioning cells that stay empty and makes the test more conservative, so the evidence that the phase drives the excess is clearest at $k \leq 3$. At $k = 7$ to 10 the placebo alone removes much of the rejection rate (at $k = 10$, from 21.4 to 5.5 per cent), which is what unvisited states produce: a threshold that rises steeply with the degrees of freedom while the statistic barely moves. A long-history target carries its phase, and the unvisited states make the test conservative.

Before the calibration, the thesis predicted that shared periodic drive would make the analytic test reject too often, from the effective-sample-size approximation

$$N_{\text{eff}} \approx \frac{N}{1 + 2\sum_{j \geq 1} \rho_j}, \tag{S.6}$$

under which G would be inflated by about N/N_{eff}, most for the most periodic units. The heuristic was mis-specified: it comes from the variance of a single series' mean, not from dependence between two series; a strictly phase-locked change indicator has an autocorrelation that alternates in sign and does not decay, so the sum does not converge; and the two-series correction of Cliff et al. (2021) is computed from residuals that, for transfer entropy, already include the target's past. Every test also uses nominal degrees of freedom, whereas on deterministic data those attained over the occupied cells are far smaller (Panzeri and Treves, 1996; Panzeri et al., 2007); the placebo result is its measured symptom.

S.3.1.8 Random Boolean circuits: design and further results

Each circuit has 40 nodes: five persistent external inputs, a clock node alternating every step, and 34 gates of eight types (46 per cent inverting), each reading up to three earlier nodes, with feedback. Every gate updates from its parents' previous values, and every node is flipped with probability 0.02 at every step, so every joint state is reachable. Series of 1,000 to 30,000 steps in 20 replicates are recorded as states and as rising edges. Table S8 gives the recovery at 10,000 steps that Section 4.6 summarises. XOR edges are recovered at 0.94 to 0.98 on states, so synergy did not defeat IDTxl's greedy selection here. On rising edges, recall is lost on inverting gates, whose input edges are detected at 0.63 against 0.98 for non-inverting gates, and false positives grow with length, because the parents' rising-edge series need not screen a node off from its other ancestors. Thresholded by the test, pairwise scores give dense networks that worsen with data, not through test errors, since on the pairs that are independent by construction the

TABLE S8: On state data, multivariate inference recovers the wiring of random Boolean circuits almost without error; on rising edges its precision falls to 0.39, and significance-thresholded bivariate networks are dense under both representations. Recovery at 10,000 steps (40 nodes; 20 replicates for pairwise methods, 10 for IDTxl). Means with 95% bootstrap intervals over replicates. TE, transfer entropy; MCC, Matthews correlation coefficient; FPR, false-positive rate; top-$|E|$, the pairwise ranking cut at the number of true edges. Rows for pairwise TE thresholded by Benjamini–Hochberg control are omitted: the corrected values have no surviving source file (Section S.3.1.9).

Method	Representation	Precision	Recall	MCC	FPR
IDTxl multivariate TE	state	0.98 [0.97, 0.99]	1.00 [0.99, 1.00]	0.99 [0.98, 1.00]	0.001
IDTxl multivariate TE	rising	0.39 [0.36, 0.42]	0.86 [0.81, 0.89]	0.55 [0.51, 0.59]	0.055
IDTxl bivariate TE	state	0.08 [0.08, 0.09]	0.99 [0.97, 1.00]	0.21 [0.20, 0.22]	0.449
IDTxl bivariate TE	rising	0.10 [0.09, 0.11]	0.99 [0.97, 1.00]	0.25 [0.23, 0.26]	0.364
Pairwise TE, top-$\|E\|$	state	0.68 [0.66, 0.70]	0.68 [0.66, 0.70]	0.67 [0.65, 0.69]	0.013
Pairwise TE, top-$\|E\|$	rising	0.49 [0.45, 0.52]	0.49 [0.45, 0.52]	0.47 [0.43, 0.50]	0.021

test rejects in 5.7 to 6.3 per cent of tests, but through dependent two-step and shared-parent pairs. The signed effect δ is negative for all 427 inverting edges under both representations.

S.3.1.9 A defect in the pairwise significance pipeline

The first run of the Boolean benchmark reported that pairwise Benjamini–Hochberg networks lost recall as series lengthened while the same scores ranked true edges near the top at every length, which is impossible for a p-value monotone in the estimate. The thesis's own correction routine treated a log p-value of $-\infty$, returned when the statistic underflows, as missing, and so removed the most significant tests. The synthetic pairwise networks were recomputed for all 160 replicate runs; the recording's main analysis had replaced underflowed values by a finite value below every other, and recomputing its networks with exact log-tail p-values gives networks of the same sizes. The final run's file of synthetic results (`synthetic_summary.csv`) holds only the values before this correction. The corrected pairwise values have no surviving source file, so Tables S8 omit them.

S.3.1.10 The simulator, the simulated chip and the NMOS circuits

A Python port of the Visual6502 switch-level algorithm, written for the thesis, supplies the synthetic NMOS circuits, the simulated chip and its interventional reference. It is validated by execution: a run is accepted only if every byte the simulated processor writes matches an independent reference. Program 1 writes the first 32 Fibonacci numbers modulo 256 and the output of an 8-bit shift register to memory (60,000 half-steps); it does not exercise subtraction, the stack, subroutine calls or indirect addressing. Program 2, written for the held-out tests, exercises those: each iteration fills a 16-byte table from a linear congruential sequence, bubble-sorts it and calls a checksum subroutine that subtracts through an indirect pointer. Its output was checked against an independent 6502 emulator (py65 1.2.0) running the same binary, and a deliberately altered byte failed the check. The check cannot detect a node error that does not change what is written, and the simulator was never compared node by node with the reference simulator.

The simulator runs the full netlist, recording all 1,702 nodes after every half-step; 1,100 change at least once. Three representations are derived from each node's state: the state, the rising edge, as the recording was described, and the change, as the recording most probably is, giving 744, 682 and 495 groups. Nodes play the part of transistors, nodes with identical series are grouped as on the recording,

TABLE S9: Recorded as states, the simulated 6502's wiring is ranked well by lag-0 CMI and below chance by lag-1 TE; under change coding, the recording's representation, neither lag ranks the wiring well. Node-level AUC of each score against the wiring and against the interventional references within the same half-step (E_0) and after the next (E_1), with 95% target-cluster bootstrap intervals (500 draws). In parentheses, the number of groups. Histories selected by AIS up to $k = 6$. MI, mutual information; CMI, conditional mutual information; TE, transfer entropy; AIS, active information storage; activity baseline, the product of the two nodes' change rates. State and rising-edge rows: H2's pre-specified replication. †Supplementary: change-of-state rows, intervals and baseline.

Representation	Score	Wiring	Same half-step (E_0)	Next half-step (E_1)
State (744)	Zero-lag MI	0.695 [0.676, 0.719]	0.600 [0.590, 0.613]	0.604 [0.594, 0.617]
	Lag-0 CMI	0.749 [0.733, 0.765]	0.711 [0.698, 0.725]	0.673 [0.660, 0.689]
	Lag-1 TE	0.431 [0.410, 0.450]	0.677 [0.662, 0.692]	0.671 [0.657, 0.686]
	Activity baseline	0.592 [0.579, 0.607]	0.649 [0.627, 0.664]	0.605 [0.581, 0.623]
Rising edge (682)	Zero-lag MI	0.604 [0.589, 0.620]	0.645 [0.628, 0.661]	0.593 [0.574, 0.610]
	Lag-0 CMI	0.581 [0.562, 0.598]	0.628 [0.613, 0.642]	0.614 [0.600, 0.631]
	Lag-1 TE	0.554 [0.537, 0.570]	0.654 [0.641, 0.668]	0.601 [0.587, 0.619]
	Activity baseline	0.591 [0.576, 0.607]	0.645 [0.626, 0.663]	0.603 [0.580, 0.622]
Change of state (495)	Zero-lag MI	0.477 [0.449, 0.509]	0.575 [0.556, 0.600]	0.603 [0.584, 0.627]
	Lag-0 CMI	0.520 [0.495, 0.550]	0.573 [0.548, 0.603]	0.648 [0.630, 0.670]
	Lag-1 TE	0.374 [0.347, 0.403]	0.622 [0.600, 0.645]	0.619 [0.599, 0.641]
	Activity baseline	0.391 [0.366, 0.422]	0.587 [0.565, 0.612]	0.635 [0.616, 0.658]

and the *node level* is the analogue of the physical level. The wiring reference is the node image of the terminal-to-gate relation. For the interventional references the chip state was saved at 40 random moments; for each non-constant node the chip was restored, the node clamped to the opposite value and re-settled, and $E_0[i, j]$ and $E_1[i, j]$ record the fraction of moments at which node j differs within the same half-step and after the next. Of the 3,154 wired pairs whose source was clamped, 63.9 per cent show an effect within the same half-step, and of the 57,084 causal pairs under E_0 only 3.5 per cent are wired. The ordered pairs of intervened nodes divide into 55,110 *immediate*, 84,908 *delayed-only* and 938,220 pairs with *no* effect. Each NMOS circuit has 80 gates fed by six external inputs and 16 dynamic latches clocked by two non-overlapping phases, simulated for 2,000 to 60,000 half-steps in 10 replicates. Intervals on the simulated chip are 95 per cent target-cluster bootstrap intervals (500 draws).

Table S9 gives the scores that Section 4.6 summarises. Under rising edges lag 1 is the better score on the NMOS circuits at every length, and on the chip lag 0 leads by only $+0.027$. Under change coding, chosen as the relevant replication after the inverter test had indicated the recording's probable coding, lag-0 CMI keeps an AP lift of 3.97, close to the recording's; the discrimination fails largely because lag-1 transfer entropy falls below chance. Against same-half-step influence the activity baseline reaches 0.649, beside 0.677 for lag 1 and 0.711 for lag 0. In the multivariate runs of Section 4.6, the 30 chip targets were drawn at random among those with a wired input; on the NMOS circuits admitting lag 0 also multiplies recall four- to six-fold under states, and under change coding (four NMOS circuits, supplementary) 288 of the 306 source variables selected with lags 0 and 1 are at lag 0, and recall rises to 0.399 at a precision of 0.064: lag-0 candidates carry wiring, as under states, but also much dependence that is not wiring. IDTxl was not run on the chip under change coding.

S.3.1.11 Clamping the channel nodes of pass transistors

A supplementary test clamped one channel node of each of 504 pass transistors, 190 of them clock-gated, and asked whether the other changed. It did within the same half-step for 94.4 per cent, and for all 190 clock-gated ones, against 84.7 per cent for wired gate-to-channel pairs and 9.0 per cent for random pairs. The result largely follows from the simulation rule: a conducting pass transistor joins its channel nodes into one node group, so clamping one channel node sets the other whenever the gate is on, and a clock-gated transistor is on every other phase. The test confirms the simulator's semantics and is not independent evidence of signal flow; it also shows, before A1 was designed, that a clamp applied while a latch is open crosses it within the half-step, the fact that A16 later used (Section 5.6.1).

TABLE S10: Alternatives to the multivariate transfer entropy used here, none of which was run: when each reads the source, whether the bound of Proposition 2 applies to it, and whether running it could change the conclusions. Graph-algorithmic post-processing of bivariate networks (Wollstadt et al., 2015) separates direct from indirect interactions using reconstructed delays, which for this circuit's wiring are zero; the remedy of Park et al. (2023) for synchrony assumes a continuous-time model class that switch-level logic does not fit; and subtracting an average stimulus-locked response before estimating Granger causality leaves a shared component wherever the response varies from trial to trial (Wang et al., 2008), which is why conditioning on the phase here removes the clock's average effect and leaves its variation from cycle to cycle.

Method	Reads the source	Bound of Proposition 2	Could it change the conclusions?
Point-process GLM (Pillow et al., 2008); kinetic Ising (Roudi and Hertz, 2011)	One step earlier	Yes, for a model saturated in the conditioning set (consequence 3)	No for same-phase wiring; it could rank latch pairs as lag-1 TE does
PCMCI, lagged links (Runge et al., 2019; Runge, 2018)	At lags of one or more	Yes	No for same-phase wiring
PCMCI+ (Runge, 2020); SVAR-LiNGAM (Hyvärinen et al., 2010), contemporaneous links	In the same sample	No	Possibly for within-phase wiring, but faithfulness and noise assumptions fail for noiseless logic
Graph post-processing of bivariate TE (Wollstadt et al., 2015)	Reconstructed delays	As bivariate TE	No: the wiring's delays are zero
Convergent cross mapping (Sugihara et al., 2012)	Lagged coordinates of an attractor	Not applicable	No attractor can be reconstructed from a binary series sampled once per phase
Optimal causation entropy (Sun et al., 2015)	Lagged states	Yes	No for same-phase wiring
Ensemble, phase-resolved TE (Gómez-Herrero et al., 2015; Wollstadt et al., 2014)	Within one phase across cycles	Removed by construction (consequence 4)	Partly done here, by conditioning on parity
Spectrally resolved TE (Pinzuti et al., 2020)	By frequency band	Not applicable	The clock lies at the Nyquist frequency; conditioning on the phase is more direct

S.4 Supporting table for Chapter 3

TABLE S11: Record of the literature search reported in Section 3.5, carried out on 1 November 2026. Inclusion criterion, fixed before screening: a study that applies a directed or information-theoretic inference method to activity recorded from a designed digital circuit and scores the inferred structure against the circuit's wiring or an interventional reference. Records: works listed (citation index) or relevant results inspected (web). Retained: studies meeting the criterion. The web searches used a general web search engine with the queries exactly as shown.

Step	Source and query	Records	Retained
Citation screen	Semantic Scholar Graph API, works citing DOI 10.1371/journal.pcbi.1005268 (Jonas and Kording 2017), paged in blocks of 100. Titles screened for all records; abstracts retrieved for the 95 records in the blocks that contained candidate titles	318 records (the index's citation count), including some duplicate records	1: Wang and Kording (2023), listed twice (arXiv and TMLR records)
Web search 1	`"transfer entropy" microprocessor OR "MOS 6502" OR "digital circuit" connectivity inference netlist`	10	1 borderline: Tehrani-Saleh and Adami (2020)
Web search 2	`"Granger causality" "6502" microprocessor transistor connectivity`	10	Wang and Kording (2023) (repeat)
Web search 3	`"transfer entropy" "Boolean network" inference of network structure from time series known topology`	10	0
Web search 4	`"causal discovery" "digital circuit" OR "logic circuit" time series ground truth benchmark`	10	Wang and Kording (2023) (repeat)
Web search 5	`"directed information" OR "transfer entropy" "Boolean network" inference wiring accuracy deterministic`	10	0
Web search 6	`"netlist" "transfer entropy" OR "Granger causality" OR "mutual information" reverse engineering circuit from activity traces`	10	Borderline study (repeat)
Web search 7	`"directed information" OR "Granger causality" "logic gates" OR "digital circuit" infer connectivity from simulated activity known wiring`	10	Borderline study (repeat)
Web search 8	`"transfer entropy" OR "causal discovery" "microprocessor" activity "ground truth" connectivity 2024 OR 2025 OR 2026`	10	Both (repeats)
Total			1 study, and 1 borderline study whose circuits are evolved rather than designed

S.5 Literature search

TABLE S12: The two retained studies and the nearest neighbours found by the search or by the review, with the respect in which each differs from this thesis. Found by: C, the citation screen; W, the web searches; R, the review of Chapter 3. Source: the search of Table S11 and the publications cited.

Study	System, method and reference	Found	How it differs from this thesis
Jonas and Kording (2017)	MOS 6507 in a simulated Atari 2600; conditional Granger causality on five Gaussian-weighted ($\sigma = 500\,\mu$m), low-pass-filtered LFP-like signals at sites in functional subcircuits	–	The precedent itself: directed analysis at the resolution of a local field potential, by design, and not scored against any reference
Wang and Kording (2023)	MOS 6502 transistors at simulator update steps; correlation, MI, linear Granger causality, VAR-LiNGAM and a learned method; perturbation reference in windows of 128 half-clocks	C, W	Retained. No transfer entropy and no multivariate information-theoretic inference; the netlist set aside as lacking direction; short windows; its aim is supervised learning of discovery procedures
Tehrani-Saleh and Adami (2020)	Evolved circuits of 4–17 deterministic two-input gates; pairwise TE against a ground truth from connections and logic tables	W	Borderline. Circuits evolved rather than designed, small, and updated with a one-step delay, so lag 1 is the correct delay by construction
Tian et al. (2019)	Hodgkin–Huxley, Lorenz and logistic-map networks with binarised output; pairwise TE against the generating adjacency	W	"Digital" refers to binarised signals; no logic circuit
Goodman and Porfiri (2020)	Boolean network model of policy diffusion and tent-map networks; analytic TE against link weights	W	Model networks (a Boolean model of policy diffusion and chaotic maps); an analytic study of the error of TE inference
Venkatesh et al. (2020)	Graphs of clocked computational nodes; a definition of information flow by conditional mutual information	C	Theoretical definition that presupposes the graph; no inference from recorded activity
Lindsay and Bau (2023)	Artificial neural networks as test beds for neuroscience methods	C	Argues for testing methods on systems with known mechanism, using neural networks rather than digital circuits
Liang et al. (1998)	Random synchronous Boolean networks; mutual information over input sets (REVEAL) against wiring	R	Random wiring, one-step updates, transitions sampled across the state space
Lizier et al. (2011)	Random Boolean networks; information storage and transfer	R	Describes computation; infers no network and scores none

TABLE S13: The mechanisms of the framework (Section 3.7) and the analyses that test them. †Added after the main results had been seen. Test identifiers A1–A19 and R1 are the hash-logged tests of Table 4.2.

Mechanism	Tested by
1 Delay mismatch and phase confounding	H1, H2, H6; †Propositions 2 and 3 pair by pair (A3, A9) and their consequences (A12, A14, A18); †the tests of latch transfer (A1, A1b, A2, A5–A7, A13, A16, A19); †synthetic networks (A11, A15)
2 Inversion and the event representation	H4, stated when the recording was believed to mark switches to on; recall and orientation by edge class; the three representations of the simulated chip; †state against event coding (A15)
3 Shared periodic drive (3a) and indirect paths (3b)	H5; scores by relation class; the surrogate calibration; †conditioning on the phase (A4); †the clock-aligned rerun (A17)
4 Synergy in logic gates	No hypothesis; descriptive analyses
5 Wiring is not influence	The observability ceiling; the interventional references (H7); †the tests of latch transfer, including timed clamps (A16)

S.6 Supporting table for Chapter 4

TABLE S14: †Fraction of synthetic edges with negative signed influence by gate type of the target (and clock sources), at 10,000 steps; each edge is counted once (Section 4.6).

Representation	Source class	Fraction negative	Edges
rising	AND	0.046	196
rising	BUF	0	64
rising	MAJ3	0.028	106
rising	NAND	1	146
rising	NOR	1	176
rising	NOT	1	105
rising	OR	0.019	157
rising	XOR	0.435	85
rising	clock source	0.476	189
state	AND	0	196
state	BUF	0	64
state	MAJ3	0.009	106
state	NAND	1	146
state	NOR	1	176
state	NOT	1	105
state	OR	0	157
state	XOR	0.529	85
state	clock source	0.513	189

TABLE S15: †Synthetic circuits at 10,000 steps. Rate at which each IDTxl network declares indirect (two-step and shared-parent) pairs and unrelated pairs, and their ratio (Section 4.6).

Method	Representation	Indirect rate	Unrelated rate	Enrichment
IDTxl bivariate TE	rising	0.740	0.299	2.474
IDTxl bivariate TE	state	0.896	0.388	2.310
IDTxl multivariate TE	rising	0.184	0.030	6.182
IDTxl multivariate TE	state	0.001	7.53e-04	1.492

S.7 Validation

TABLE S16: Circular-shift rejection rate (per cent) of the analytic test by history band, with three conditioning sets† (the target's past only, as in the main run; the past and a placebo bit; the past and sample parity), at the nominal level and at the per-test levels of the recording's Benjamini–Hochberg network and of Bonferroni correction. Bands: the 40-target sample (2, 8, 7, 7 and 16 targets). Last column: all 54 targets with $k \leq 3$ (placebo not run). Null-mean ratio: median over targets of the empirical over the analytic null mean. Weighting each band by its share of the 442-target frame gives 20.3 per cent for the past-only test at $\alpha = 0.05$, and by its share of all 519 targets 24.2 per cent.

		History band k					40-target sample		All 54,
Conditioning set	Threshold	1–3	4–6	7–9	10	11–12	Mean	Frame-weighted	$k \leq 3$
Target past only	$\alpha = 0.05$	58.56	38.91	34.33	21.43	0.16	20.53	20.26	36.99
	BH level, 0.0167	54.23	37.18	33.58	20.82	0.15	19.73	19.47	33.12
	Bonferroni, 1.9×10^{-7}	39.22	29.78	28.72	17.24	0.094	16.00	15.79	17.83
	Median null-mean ratio	20.63	3.40	1.14	0.63	0.13	0.63	–	–
Past and a placebo bit	$\alpha = 0.05$	53.66	30.46	20.46	5.50	0	13.32	13.12	–
	BH level, 0.0167	50.68	29.50	19.91	5.23	0	12.83	12.64	–
	Bonferroni, 1.9×10^{-7}	37.31	23.86	17.31	3.70	0	10.32	10.16	–
	Median null-mean ratio	10.44	1.72	0.57	0.32	0.07	0.32	–	–
Past and sample parity	$\alpha = 0.05$	8.57	1.86	0.59	0.006	0	0.91	0.89	3.85
	BH level, 0.0167	6.40	1.64	0.54	0.005	0	0.74	0.73	2.83
	Bonferroni, 1.9×10^{-7}	0.90	0.79	0.33	0.003	0	0.26	0.26	0.65
	Median null-mean ratio	0.71	0.13	0.09	0.03	0.04	0.05	–	–

S.7.1 Supporting tables for Section 4.6

TABLE S17: IDTxl multivariate inference with sources at lag 1 only or at lags 0 and 1. Panel A: the simulated 6502, 30 target groups; edges are group pairs, and precision (95% target-cluster interval), recall and MCC are over the node pairs they represent. Pairwise: precision of the pairwise lag-1 network of the same size on the same targets. E_0, E_1: precision against the interventional references. Panel B: NMOS circuits, node level, mean over four circuits (standard deviation of precision). †Supplementary, 2,000 half-steps only; two of the four lag-1 networks were empty, so precision is the mean over the other two.

A. Simulated 6502

Representation	Sources	Edges	Wired	Precision	Recall	MCC	Pairwise	E_0	E_1
State	Lag 1	90	0/173	0.000 [0.000, 0.000]	0.000	−0.003	0.005	0.36	0.49
	Lags 0 and 1	40	16/83	0.193 [0.110, 0.266]	0.205	0.197	0.000	0.43	0.54
Rising edge	Lag 1	99	0/247	0.000 [0.000, 0.000]	0.000	−0.003	0.014	0.28	0.39
	Lags 0 and 1	85	2/213	0.009 [0.000, 0.025]	0.023	0.012	0.012	0.23	0.35

B. NMOS circuits

		2,000 half-steps			6,000 half-steps		
Representation	Sources	Precision	Recall	MCC	Precision	Recall	MCC
State	Lag 1	0.092 (0.041)	0.081	0.066	0.102 (0.017)	0.125	0.090
	Lags 0 and 1	0.208 (0.056)	0.483	0.291	0.204 (0.057)	0.478	0.286
Rising edge	Lag 1	0.087 (0.025)	0.075	0.060	0.095 (0.028)	0.101	0.076
	Lags 0 and 1	0.018 (0.008)	0.055	−0.010	0.024 (0.012)	0.098	−0.001
Change of state†	Lag 1	0.097 (0.083)	0.031	0.030	–	–	–
	Lags 0 and 1	0.064 (0.042)	0.399	0.116	–	–	–

TABLE S18: †Lag-0 and lag-1 transfer entropy at the AIS-selected histories and at common histories $k = 1, 2, 4, 8$ for every target, on the reduced recording. Intervals: 1,000 target-cluster bootstrap draws.

Score	Physical AUC	Physical AP lift	Group AUC	Group AP lift	Within-target AUC
lag 1, selected k	0.500 [0.411, 0.596]	1.27 [0.99, 1.84]	0.469 [0.445, 0.493]	1.22 [1.04, 1.60]	0.434 [0.418, 0.450]
lag 0, selected k	0.593 [0.504, 0.683]	4.35 [2.74, 7.76]	0.632 [0.611, 0.651]	4.72 [3.76, 6.08]	0.632 [0.613, 0.651]
lag 1, k = 1	0.503 [0.415, 0.603]	1.76 [1.03, 3.17]	0.487 [0.467, 0.508]	1.40 [1.05, 2.04]	0.467 [0.451, 0.484]
lag 0, k = 1	0.585 [0.500, 0.672]	4.99 [2.43, 9.75]	0.609 [0.591, 0.627]	4.01 [3.09, 5.26]	0.620 [0.601, 0.639]
lag 1, k = 2	0.500 [0.409, 0.600]	1.16 [0.90, 2.17]	0.473 [0.450, 0.496]	1.15 [0.96, 1.69]	0.442 [0.426, 0.457]
lag 0, k = 2	0.589 [0.502, 0.678]	2.82 [1.88, 5.38]	0.618 [0.599, 0.636]	3.49 [2.79, 4.63]	0.618 [0.598, 0.638]
lag 1, k = 4	0.500 [0.410, 0.599]	1.33 [0.98, 2.12]	0.477 [0.453, 0.500]	1.21 [1.03, 1.64]	0.448 [0.432, 0.463]
lag 0, k = 4	0.592 [0.506, 0.681]	2.96 [2.02, 5.15]	0.617 [0.597, 0.638]	3.95 [3.10, 5.15]	0.618 [0.597, 0.638]
lag 1, k = 8	0.497 [0.410, 0.595]	1.26 [0.98, 1.92]	0.470 [0.446, 0.495]	1.21 [1.04, 1.56]	0.435 [0.419, 0.451]
lag 0, k = 8	0.592 [0.504, 0.682]	3.79 [2.41, 6.54]	0.625 [0.605, 0.646]	4.27 [3.39, 5.45]	0.625 [0.605, 0.645]

TABLE S19: †Groups classified by the parity of the samples in which they have events (136 even only, 309 odd only, 74 mixed; threshold 95%), and how wiring is distributed over pairs of these classes.

Pair class	Group pairs	Wired group pairs	Wired (%)	Wired physical pairs
same parity	113,532	770	0.68	1,253
opposite parity	84,048	30	0.04	34
involves a mixed group	71,262	637	0.89	2,256

TABLE S20: †Scores conditioned additionally on sample parity (and, in one row, on the quartile of total activity in the previous sample). Intervals: 1,000 target-cluster bootstrap draws.

Score	Physical AUC	Physical AP lift	Group AUC	Group AP lift	Within-target AUC
lag 0	0.593 [0.504, 0.683]	4.35 [2.74, 7.76]	0.632 [0.611, 0.651]	4.72 [3.76, 6.08]	0.632 [0.613, 0.651]
lag 0 | parity	0.633 [0.561, 0.704]	4.50 [2.87, 7.95]	0.694 [0.678, 0.708]	4.76 [3.78, 6.08]	0.711 [0.694, 0.727]
lag 0 | parity, global state	0.624 [0.555, 0.692]	6.12 [4.03, 10.48]	0.664 [0.649, 0.678]	5.04 [4.03, 6.56]	0.660 [0.640, 0.680]
lag 1	0.500 [0.411, 0.596]	1.27 [0.99, 1.84]	0.469 [0.445, 0.493]	1.22 [1.04, 1.60]	0.434 [0.418, 0.450]
lag 1 | parity	0.547 [0.487, 0.609]	1.32 [1.07, 1.85]	0.488 [0.466, 0.512]	1.20 [1.04, 1.54]	0.490 [0.471, 0.507]
MI	0.545 [0.498, 0.591]	3.21 [2.02, 5.62]	0.597 [0.578, 0.615]	3.38 [2.73, 4.37]	0.615 [0.594, 0.636]
MI | parity	0.632 [0.596, 0.671]	4.17 [2.62, 7.25]	0.737 [0.722, 0.750]	4.65 [3.82, 5.85]	0.767 [0.755, 0.779]

S.8 Pairwise transfer entropy

Orientation and conditional entropy (Section 5.8). For lag-0 CMI, writing Y^- for $Y_t^{(k)}$ and X^- for $X_t^{(k)}$,

$$\begin{aligned} I(X_t; Y_t \mid Y^-) - I(Y_t; X_t \mid X^-) = &\big[H(Y_t \mid Y^-) - H(X_t \mid X^-)\big] \\ &- \big[H(Y_t \mid X_t, Y^-) - H(X_t \mid Y_t, X^-)\big], \end{aligned} \tag{S.7}$$

TABLE S21: †Lag-0 CMI group AUC and the H2 contrast (lag-0 CMI minus lag-1 TE in AUC) in ten consecutive blocks of the reduced recording, with the main run's histories and each sample's history read from the full series (95% target-cluster intervals). Block 1 contains the misaligned samples 1–1,407.

Block	Samples	Lag-0 CMI, group AUC	Δ AUC, group	Δ AUC, physical
1	0–13,392	0.684	+0.173 [+0.150, +0.195]	+0.097 [+0.061, +0.139]
2	13,393–26,785	0.536	+0.116 [+0.099, +0.133]	+0.065 [+0.046, +0.089]
3	26,786–40,178	0.525	+0.097 [+0.081, +0.114]	+0.055 [+0.036, +0.078]
4	40,179–53,571	0.574	+0.182 [+0.159, +0.204]	+0.100 [+0.074, +0.132]
5	53,572–66,964	0.528	+0.094 [+0.079, +0.111]	+0.056 [+0.037, +0.078]
6	66,965–80,357	0.560	+0.163 [+0.142, +0.186]	+0.090 [+0.066, +0.119]
7	80,358–93,749	0.555	+0.152 [+0.132, +0.172]	+0.083 [+0.061, +0.109]
8	93,750–107,141	0.526	+0.097 [+0.081, +0.113]	+0.055 [+0.035, +0.078]
9	107,142–120,533	0.576	+0.182 [+0.159, +0.205]	+0.099 [+0.073, +0.132]
10	120,534–133,925	0.528	+0.095 [+0.079, +0.111]	+0.056 [+0.037, +0.078]

TABLE S22: †Orientation of the 1,172 wired group pairs whose source is not the clock group, by lag-1 transfer-entropy asymmetry, against the reversed event-rate rule (as Table S30). Target: target-cluster interval; two-way: sources and targets resampled.

History	Level	Accuracy [target]	Rate rule	TE − rate rule [target]	Two-way
$k = 1$	group	0.538 [0.498, 0.577]	0.632	−0.094 [−0.147, −0.037]	[−0.194, 0.023]
$k = 1$	physical	0.535 [0.487, 0.582]	0.590	−0.054 [−0.118, 0.012]	[−0.172, 0.076]
$k = 2$	group	0.461 [0.421, 0.503]	0.632	−0.171 [−0.238, −0.094]	[−0.322, −0.001]
$k = 2$	physical	0.490 [0.440, 0.541]	0.590	−0.099 [−0.186, −0.005]	[−0.259, 0.076]
$k = 4$	group	0.531 [0.492, 0.572]	0.632	−0.101 [−0.168, −0.031]	[−0.232, 0.040]
$k = 4$	physical	0.556 [0.504, 0.603]	0.590	−0.034 [−0.117, 0.052]	[−0.178, 0.131]
$k = 8$	group	0.629 [0.590, 0.672]	0.632	−0.003 [−0.060, 0.059]	[−0.100, 0.107]
$k = 8$	physical	0.612 [0.562, 0.659]	0.590	+0.023 [−0.049, 0.099]	[−0.094, 0.150]
AIS-selected k	group	0.665 [0.627, 0.705]	0.632	+0.033 [−0.017, 0.087]	[−0.043, 0.124]
AIS-selected k	physical	0.632 [0.587, 0.678]	0.590	+0.043 [−0.019, 0.108]	[−0.054, 0.163]

since $I(X_t; Y_t \mid Y^-) = H(Y_t \mid Y^-) - H(Y_t \mid X_t, Y^-)$ and likewise in reverse. The first bracket is the conditional-entropy predictor of H3, and lag-1 transfer entropy, $H(Y_t \mid Y^-) - H(Y_t \mid Y^-, X_{t-1})$, contains the same first term, so both asymmetries favour the end of an edge that is less predictable from its own past.

TABLE S23: †The pre-specified cross-validated logistic test of H3 on the 1,437 wired group pairs: out-of-fold AUC of the nuisance predictors alone and the increment from adding the lag-1 TE asymmetry, as run (unscaled features) and refitted with standardised features, weighted by physical edges (pre-specified) or once per group pair. Intervals: target-cluster bootstrap of the out-of-fold score.

Model	AUC	Increment	Interval
unscaled (as run); physical-edge weighted (pre-specified)			
Nuisance predictors only	0.826 [0.774, 0.870]	–	–
+ TE asymmetry, saved target-specific k	–	−0.0085	[−0.0309, +0.0028]
+ TE asymmetry, common $k = 1$	–	−0.0087	[−0.0316, +0.0036]
+ TE asymmetry, common $k = 2$	–	−0.0074	[−0.0297, +0.0039]
+ TE asymmetry, common $k = 4$	–	−0.0085	[−0.0313, +0.0032]
+ TE asymmetry, common $k = 8$	–	−0.0094	[−0.0323, +0.0024]
standardised, C = 1; physical-edge weighted (pre-specified)			
Nuisance predictors only	0.871 [0.832, 0.904]	–	–
+ TE asymmetry, saved target-specific k	–	−0.0154	[−0.0560, +0.0057]
+ TE asymmetry, common $k = 1$	–	−0.0111	[−0.0369, +0.0019]
+ TE asymmetry, common $k = 2$	–	−0.0090	[−0.0344, +0.0039]
+ TE asymmetry, common $k = 4$	–	−0.0109	[−0.0399, +0.0041]
+ TE asymmetry, common $k = 8$	–	−0.0170	[−0.0566, +0.0038]
unscaled (as run); group pairs, unit weight			
Nuisance predictors only	0.721 [0.691, 0.751]	–	–
+ TE asymmetry, saved target-specific k	–	+0.0164	[+0.0112, +0.0211]
+ TE asymmetry, common $k = 1$	–	+0.0065	[+0.0017, +0.0118]
+ TE asymmetry, common $k = 2$	–	+0.0085	[+0.0034, +0.0135]
+ TE asymmetry, common $k = 4$	–	+0.0158	[+0.0106, +0.0209]
+ TE asymmetry, common $k = 8$	–	+0.0185	[+0.0125, +0.0239]
standardised, C = 1; group pairs, unit weight			
Nuisance predictors only	0.803 [0.773, 0.832]	–	–
+ TE asymmetry, saved target-specific k	–	+0.0252	[+0.0117, +0.0375]
+ TE asymmetry, common $k = 1$	–	+0.0065	[+0.0003, +0.0131]
+ TE asymmetry, common $k = 2$	–	+0.0071	[+0.0002, +0.0143]
+ TE asymmetry, common $k = 4$	–	+0.0115	[+0.0032, +0.0204]
+ TE asymmetry, common $k = 8$	–	+0.0204	[+0.0084, +0.0314]

TABLE S24: †(Section 5.7) Lag-0 CMI under the controls called for by the rivals to H2. The primary test, the common-history row and the rates-only model are from the main run; all other rows are supplementary. Panel A: discrimination. Panel B: the paired AUC contrast with lag-1 transfer entropy under the same control. "| parity": conditioned on sample parity (in panel B, lag 1 is conditioned in the same way). Global state: quartile of total activity in the previous sample. Clock group excluded: as source and target. Rate strata: AUC within cells of source-rate decile by target-rate decile. Rates-only model: cross-validated logistic model of the two log rates, folds by target. Node level: each connection between active gate nodes counted once; its node-level AUC, AP lift and within-target-node AUC are placed in the group columns.

A. Discrimination

Control	Physical AUC	Group AUC	Group AP lift	Within-target AUC
None (primary test)	0.593 [0.504, 0.683]	0.632 [0.611, 0.651]	4.72 [3.76, 6.08]	0.632 [0.613, 0.651]
Clock-aligned	0.527 [0.441, 0.614]	0.561 [0.538, 0.582]	4.62 [3.67, 6.00]	0.511 [0.479, 0.541]
Common history $k = 1$	0.585 [0.500, 0.672]	0.609 [0.591, 0.627]	4.01 [3.09, 5.26]	0.620 [0.601, 0.639]
| parity	0.633 [0.561, 0.704]	0.694 [0.678, 0.708]	4.76 [3.78, 6.08]	0.711 [0.694, 0.727]
| parity and global state	0.624 [0.555, 0.692]	0.664 [0.649, 0.678]	5.04 [4.03, 6.56]	0.660 [0.640, 0.680]
Clock group excluded	0.641 [0.598, 0.679]	0.682 [0.656, 0.708]	6.65 [5.17, 8.70]	0.761 [0.739, 0.781]
Within rate strata	0.583 [0.469, 0.687]	0.626 [0.600, 0.651]	–	–
Node level	–	0.615 [0.594, 0.635]	3.84 [2.96, 5.64]	0.586 [0.567, 0.605]
Comparators				
Rates-only model	0.625 [0.585, 0.662]	0.623 [0.603, 0.642]	10.39 [8.54, 12.65]	0.715 [0.692, 0.736]
Source event rate alone	0.626 [0.586, 0.665]	0.633 [0.614, 0.652]	20.26 [16.29, 24.65]	0.721 [0.699, 0.742]
Rates model refitted without the clock group	0.581 [0.540, 0.618]	0.576 [0.553, 0.599]	1.24 [1.11, 1.43]	0.609 [0.587, 0.630]

B. Δ AUC, lag 0 minus lag 1

Control	Physical	Group
None (primary test)	$+0.094$ [$+0.065$, $+0.128$]	$+0.163$ [$+0.142$, $+0.184$]
Clock-aligned	$+0.087$ [$+0.064$, $+0.117$]	$+0.165$ [$+0.144$, $+0.187$]
Common history $k = 1$	$+0.082$ [$+0.057$, $+0.113$]	$+0.122$ [$+0.103$, $+0.140$]
| parity	$+0.086$ [$+0.039$, $+0.132$]	$+0.205$ [$+0.180$, $+0.229$]
Clock group excluded	$+0.169$ [$+0.133$, $+0.206$]	$+0.204$ [$+0.179$, $+0.231$]
Node level	$+0.145$ [$+0.127$, $+0.165$]	

TABLE S25: †Group-level discrimination against the directed netlist reference (as in the main analysis), against an undirected version (each unordered pair scored by the larger of its two directed values and positive if wired in either direction), and against the netlist with the 228 ordered pass-neighbour pairs added as positives in both directions (prevalence 0.535 per cent and 0.619 per cent). For a symmetric score such as MI the directed evaluation differs from the undirected one only through ties between each wired pair and its reverse twin, which counts as a negative. No intervals were computed (Sections S.8.1.6 and S.8.1.4).

	Directed netlist		Undirected netlist		Netlist plus pass-through	
Score	AUC	AP lift	AUC	AP lift	AUC	AP lift
Lag-1 TE	0.469	1.22	0.431	1.05	0.497	1.97
Lag-0 CMI	0.632	4.72	0.608	4.38	0.622	4.34
Zero-lag MI	0.597	3.38	0.598	3.38	0.585	3.04
Lag-0 CMI | parity	0.694	4.76	0.683	4.57	0.680	4.40
Zero-lag MI | parity	0.737	4.65	0.738	4.64	0.709	4.15

TABLE S26: Negative control: lagged transfer entropy on the unreduced recording (stride 1, common target history of three rows) at source delays in simulator rows (Section S.8.1.2). Upper block: whole recording; delays of 1, 2, 3 and 6 rows are from the main run (exploratory), the others supplementary. Middle block: first 15,000 rows (5,000 samples) excluded, with target-cluster intervals (supplementary). Lower block: the same, with the row offset of the target sample within its three-row block added to the conditioning set (supplementary). The unconditioned values reflect the row offset, which a source event reveals because almost every event after the opening rows falls in the first row of a block; conditioned on it, the scores fall to those of the reduced recording.

Delay (rows)	Delay (samples)	Physical AUC	Group AUC	Group AP lift	Within-target AUC
Whole recording, not conditioned on the row offset (no intervals)					
1	0.33	0.594	0.545	6.12	–
2	0.67	0.590	0.544	6.09	–
3	1	0.592	0.528	1.85	–
4	1.33	0.625	0.595	6.70	–
5	1.67	0.625	0.595	6.66	–
6	2	0.654	0.647	2.65	–
7	2.33	0.620	0.581	6.18	–
8	2.67	0.622	0.581	6.13	–
9	3	0.605	0.559	2.15	–
10	3.33	0.627	0.594	6.61	–
11	3.67	0.628	0.595	6.56	–
12	4	0.618	0.609	3.01	–
First 15,000 rows excluded, not conditioned on the row offset					
1	0.33	0.594 [0.518, 0.676]	0.543 [0.519, 0.568]	6.16 [4.97, 7.89]	0.669 [0.644, 0.693]
2	0.67	0.599 [0.518, 0.687]	0.542 [0.518, 0.567]	6.15 [4.94, 7.86]	0.670 [0.644, 0.694]
3	1	0.591 [0.505, 0.681]	0.527 [0.504, 0.551]	1.82 [1.61, 2.14]	0.641 [0.616, 0.664]
4	1.33	0.628 [0.554, 0.711]	0.593 [0.570, 0.616]	6.80 [5.60, 8.60]	0.729 [0.708, 0.749]
5	1.67	0.631 [0.554, 0.717]	0.594 [0.570, 0.617]	6.78 [5.57, 8.54]	0.730 [0.708, 0.750]
6	2	0.653 [0.583, 0.736]	0.645 [0.625, 0.666]	2.61 [2.35, 2.97]	0.774 [0.755, 0.791]
First 15,000 rows excluded, conditioned on the row offset within the block					
3	1	0.434 [0.348, 0.527]	0.404 [0.383, 0.427]	1.35 [0.99, 2.01]	0.337 [0.315, 0.360]
4	1.33	0.401 [0.328, 0.474]	0.445 [0.428, 0.463]	0.93 [0.91, 0.97]	0.426 [0.411, 0.443]
6	2	0.475 [0.388, 0.565]	0.471 [0.453, 0.489]	1.04 [0.98, 1.13]	0.423 [0.399, 0.451]

TABLE S27: AUC of lag-1 transfer entropy and lag-0 CMI with wired and unwired pairs compared only within strata†, whole recording. Upper block: pairs whose members change in samples of the same parity, of opposite parity, or that involve a mixed group, with and without the clock group (group 6); the second column gives the wired group pairs in each stratum. Lower block: comparisons only within the same decile of a property of the source group, or within cells of source-rate decile by target-rate decile; descriptive. A lag-1 AUC clearly above 0.5 at matched source properties would have contradicted the heuristic reading of Proposition 1 (Section S.8.1.1), and none occurs.

		Lag-1 TE		Lag-0 CMI	
Comparison	Wired	Group	Physical	Group	Physical
Same parity	770	0.472 [0.441, 0.503]	0.383 [0.334, 0.438]	0.601 [0.567, 0.635]	0.497 [0.446, 0.548]
Opposite parity	30	0.752 [0.653, 0.841]	0.704 [0.590, 0.802]	0.720 [0.637, 0.795]	0.662 [0.573, 0.754]
Involves a mixed group	637	0.481 [0.450, 0.512]	0.601 [0.434, 0.741]	0.533 [0.499, 0.563]	0.640 [0.469, 0.772]
clock group excluded	372	0.630 [0.585, 0.666]	0.590 [0.544, 0.636]	0.727 [0.696, 0.755]	0.719 [0.675, 0.761]
Source self-information $I(X_{t-1}; X_t)$	–	0.378 [0.355, 0.402]	0.460 [0.383, 0.546]	0.587 [0.562, 0.611]	0.557 [0.480, 0.640]
Source event rate	–	0.383 [0.358, 0.408]	0.452 [0.372, 0.543]	0.594 [0.568, 0.619]	0.542 [0.459, 0.625]
Source entropy	–	0.434 [0.410, 0.460]	0.479 [0.403, 0.565]	0.644 [0.619, 0.669]	0.584 [0.508, 0.664]
Source rate × target rate	–	0.350 [0.326, 0.376]	0.455 [0.357, 0.566]	0.626 [0.600, 0.651]	0.583 [0.469, 0.687]

TABLE S28: Scores by structural relation class (transistor-path definition, Section 4.4), whole recording. Mean scores (millibits) and the fraction declared by Benjamini–Hochberg are weighted by transistor pairs (physical). The last two columns† give the percentage of each class's group pairs within the 1,443 highest-ranked group pairs of each score, a budget equal to the size of the multivariate network of Chapter 6. The 1,443 highest-ranked lag-1 pairs include 27 of the 227 shared-parent pairs, an enrichment over unrelated pairs of 22.8 against 2.7 for wired pairs, and 3 of the 690 two-step paths.

	Pairs		Mean score (millibits, physical)			BH declared	Top-1,443 inclusion (%)	
Relation	Physical	Group	Lag-1 TE	Lag-0 CMI	MI	(%, physical)	Lag-1 TE	Lag-0 CMI
Wired (direct)	3,543	1,437	8.4	28.1	49.8	19.8	1.39	7.93
Reverse only	3,543	1,437	7.3	26.4	49.8	19.1	0.49	7.45
Two-step path	13,117	690	10.3	18.0	33.4	17.1	0.43	4.78
Shared parent	1,082	227	40.8	23.0	26.0	58.7	11.89	7.05
Shared child	7,382	1,155	4.3	27.5	46.4	23.8	0.61	7.62
Unrelated	9,969,243	263,896	6.4	10.4	16.7	26.7	0.52	0.41

S.8.1 Supporting analyses for Chapter 5

This section holds the tables and side analyses that Chapter 5 cites but does not need in full: discrimination within parity strata, scores by structural relation class, the NMOS latch tests, the H3 and H4 tables, the rival explanations for lag 0, what the thresholded lag-1 network declares, and the full list of the account's written predictions with their outcomes.

S.8.1.1 What Proposition 1 does and does not imply

Proposition 1 motivated H2, and the aliasing argument does not depend on it. The proposition concerns node states and does not carry over to change indicators (Section 2.6), so on this recording it

TABLE S29: Hash-logged tests on NMOS circuits, change coding, 6,000 half-steps: A8 on ten circuits (seeds 500 to 509) and A10 on 50 new, held-out circuits (seeds 600 to 649). Stratified AUC with 95% circuit-bootstrap intervals (1,000 draws), against pairs unrelated by wiring or latch. Of the 160 latches in A8, 21 input–storage pairs can be scored; in the others the input and storage nodes have identical change series or one never changes. A10 scores 95 input–storage pairs in 27 circuits and 481 storage–reader pairs. A8's paired differences are supplementary; A10's are hash-logged. At 20,000 half-steps every A10 value is within 0.014 of the value shown.

Relation	Score	A8, 10 circuits	A10, 50 new circuits
Latch input to storage node	Lag-1 TE	0.858 [0.731, 0.892]	0.779 [0.615, 0.873]
	Lag-0 CMI	0.800 [0.768, 0.980]	0.850 [0.707, 0.939]
	Lag 1 minus lag 0	$+0.057$ [-0.109, $+0.124$]	-0.071 [-0.201, $+0.047$]
Storage node to reader output	Lag-1 TE	0.347 [0.189, 0.545]	0.316 [0.243, 0.405]
	Lag-0 CMI	0.790 [0.738, 0.826]	0.765 [0.717, 0.811]
	Lag 0 minus lag 1	$+0.443$ [$+0.213$, $+0.604$]	$+0.449$ [$+0.343$, $+0.551$]

is a heuristic and the operative mechanism is the aliasing. Where states exist, on the simulated chip, lag-0 CMI ranks wiring at a node-level AUC of 0.749 and lag-1 transfer entropy at 0.431 (Section 4.6). At matched source properties (Table S27, lower block), a lag-1 AUC clearly above 0.5 would have contradicted the heuristic reading, and none occurs, while lag-0 CMI lies above 0.5 at the group level under every matching.

S.8.1.2 A negative control at the simulator's resolution

The main run also computed lagged transfer entropy on the unreduced recording, three simulator rows per phase, where the order of switching within a phase would be visible. Read without further conditioning, it appears to recover the wiring: with the first 15,000 rows excluded, a delay of six rows gives a group AUC of 0.645 (Table S26). The gain is an artefact of row alignment. After the opening rows almost every event falls in the first row of its three-row block, so a source event at a given delay tells the estimator which row of the block the target sample occupies, and so whether the target can change at all. This is the aliasing of Section 5.3.1 at the level of rows. Conditioned on the row offset within the block (supplementary), the six-row delay falls to a group AUC of 0.471. The analysis is a negative control: a lagged estimator picks up an alignment variable unless it is conditioned on. It cannot test mechanism 1, because outside the opening rows the recording holds no timing below one phase.

S.8.1.3 IDTxl's selections on the chip are not enriched for delayed influence (A1c)

On the 30 chip targets of Section 4.6, recorded as states, the 90 group-pair edges that IDTxl selects at lag 1 have a precision of 0.127 against delayed-only influence, 0.364 against immediate influence and 0 against the wiring. Pairwise lag-1 transfer entropy given the same number of sources per target reaches 0.158, 0.353 and 0 (A1c, descriptive). With lags 0 and 1 admitted, IDTxl's 40 edges reach 0.108, 0.434 and 0.193, against 0.175, 0.350 and 0 for the pairwise comparator. The prediction that conditioning would enrich the search for delayed influence was not met in either run. The comparator here allocates sources per target, so its wiring precision differs from the 0.005 of the globally cut pairwise network in Table S17.

S.8.1.4 Rival explanations for lag 0

Conditioning on the target's past. Lag-0 CMI has a higher group AUC than zero-lag MI (0.632 against 0.597). Compared within strata of source-group size, the paired difference is positive at the

TABLE S30: Orientation of wired group pairs by lag-1 transfer-entropy asymmetry at matched histories, at the group level (each group pair once) and the physical level (weighted by physical edges), among untied pairs. Rate rule: the member with the lower event rate is the target, the reverse of the pre-specified nuisance predictor, chosen after the run. Target: target-cluster interval, paired for the difference. Two-way and vertex intervals for the group differences are in Table S50; the block without the clock group's pairs is in Table S22. Against the predictors as pre-specified, computed at the physical level, the asymmetry would have read as supported: the pre-specified rate predictor orients only 0.268 of the physical weight, and the asymmetry exceeds the two conditional-entropy predictors (0.627 and 0.636) at every history. With the AIS-selected histories the asymmetry orients pass edges at 0.872 and pull-down edges at 0.654 (physical), a gap that reflects which end of an edge is more predictable from its own past (Equation (S.7)).

History	Level	Accuracy	Rate rule	TE − rate rule
$k = 1$	group	0.623 [0.589, 0.656]	0.701	-0.078 [-0.120, -0.031]
	physical	0.695 [0.642, 0.744]	0.732	-0.037 [-0.080, $+0.007$]
$k = 2$	group	0.560 [0.524, 0.600]	0.701	-0.141 [-0.198, -0.076]
	physical	0.665 [0.612, 0.714]	0.732	-0.067 [-0.125, -0.005]
$k = 4$	group	0.618 [0.582, 0.655]	0.701	-0.083 [-0.138, -0.026]
	physical	0.707 [0.658, 0.751]	0.732	-0.025 [-0.079, $+0.032$]
$k = 8$	group	0.698 [0.665, 0.735]	0.701	-0.003 [-0.050, $+0.047$]
	physical	0.744 [0.696, 0.786]	0.732	$+0.013$ [-0.035, $+0.062$]
AIS-selected k	group	0.727 [0.695, 0.759]	0.701	$+0.026$ [-0.015, $+0.070$]
	physical	0.755 [0.710, 0.797]	0.732	$+0.023$ [-0.019, $+0.069$]

group level and its interval excludes zero, whereas the physical interval includes zero (supplementary). Once both are conditioned on parity the ordering reverses: MI given parity reaches a group AUC of 0.737 and a within-target AUC of 0.767, against 0.694 and 0.711 for lag-0 CMI given parity (Table 5.1). The best pairwise score computed is therefore phase-conditioned co-activity, without the target's past. No paired interval for that difference was computed, because the score matrices it needs are not among the saved outputs.

Phase alignment. This rival holds that wired units change in the same samples because they share a phase, without either influencing the other. It accounts for most of lag 0's AUC but not for the rest: conditioning on parity raises lag 0's group AUC by 0.062 with its AP lift unchanged, and lag 0 discriminates among same-parity pairs at the group level (Table S27). Physically the rival is not excluded. Conditioning on the collective state, which also improved reconstruction in the calcium-imaging precedent (Stetter et al., 2012), matters more here than any other refinement tried. Slower rhythms, such as the instruction cycle, are controlled only through a coarse global-state variable.

Event rates. With all groups included, rates match or beat lag 0 at the group level: the rates-only model reaches an AP lift of 10.39 and a within-target AUC of 0.715 (Table S24). The group-level strength of rates comes from one hub, since 36.2 per cent of the rates model's 1,443 highest-ranked group pairs involve the clock group. Refitted without the clock group, the rates model falls to a group AUC of 0.576 while lag 0 rises to 0.682, and within cells of source-rate by target-rate decile lag 0 keeps a group AUC of 0.626.

TABLE S31: Percentage of group pairs with negative signed influence δ by edge class, within each parity stratum and overall (number of pairs in parentheses), and the pull-down-only minus unwired difference in percentage points with its target-cluster interval†. Composition-adjusted: stratum differences weighted by the pull-down-only pairs in each stratum. Group level; the physical-level overall difference is $+3.3$ [-9.1, $+15.1$]. Two-way interval of the composition-adjusted difference: [-14.6, $+0.9$]. The counts follow a recount of the edge classes, which leaves 790 pull-down-only pairs with a defined δ. Where gates act with a one-step delay and events are states or rising edges, on the synthetic circuits, δ is negative for all 427 inverting edges (Section 4.6); neither condition holds on the recording.

Stratum	Pull-down only	Pass only	Unwired	Pull-down − unwired
Same parity	86.3 (678)	82.8 (87)	91.8 (112,609)	-5.5 [-9.7, -1.4]
Opposite parity	33.3 (6)	58.3 (24)	45.8 (83,682)	-12.5 [-46.4, $+34.2$]
Involves a mixed group	50.9 (106)	31.7 (524)	57.1 (70,539)	-6.2 [-18.8, $+6.8$]
Composition-adjusted				-5.6 [-9.8, -1.7]
All strata (unstratified)	81.1 (790)	39.7 (635)	68.2 (266,830)	$+12.9$ [$+8.3$, $+17.2$]

TABLE S32: Recall within the 1,437 highest-ranked group pairs of each score (a budget equal to the number of wired group pairs): percentage of wired group pairs containing at least one edge of the class (four contain both), with target-cluster intervals†, whole recording.

Wired group pairs	Number	Lag-1 TE	Lag-0 CMI
With a pull-down edge	802	1.9 [1.0, 2.9]	11.8 [9.2, 14.8]
With a pass edge	639	0.8 [0.2, 1.7]	3.0 [1.7, 4.3]

S.8.1.5 Which edges are recovered, and what the test declares

Lag-0 CMI recovers mostly pull-down edges (Table S32). This reverses the framework's expectation that pull-down edges would be the harder class, and under change coding it is unsurprising: a pull-down source whose change flips the node it discharges changes in the same sample as its target. Part of the poor recovery of pass edges has a measured cause: the clock group, whose nearly constant series no information measure can register, is the source of 885 of the 1,614 evaluable pass edges and of 265 wired group pairs (Section 5.8). Whether the remainder are missed because the reference over-generates pass edges or because their effect depends on the value at the other channel terminal was not tested.

Thresholded, the lag-1 scores declare a dense network of dependences that are mostly not wiring. Every thresholding rule declares between 28 and 34 per cent of group pairs, at precision lifts between 0.73 and 0.78. At short histories part of this is phase dependence that a circular shift cannot remove (Section 4.6). Among the 159 targets with $k = 12$, however, whose 15 calibrated members reject 0.13 per cent of shifted pairs, Bonferroni correction still declares 23.9 per cent of pairs. False rejection is unlikely to produce those declarations, so they are mostly dependence that the shift destroys, declared at a precision close to the base rate.

TABLE S33: Hash-logged tests A11 (300 networks, 20 per condition) and A15 (80 networks with hidden co-drivers and spurious events, coupling 0.6): within-target AUC of lag-1 transfer entropy for within-sample drivers against non-drivers of the same phase, with 95% target-cluster intervals (2,000 targets and 1,000 draws per condition). Locking λ: the share of a unit's spontaneous events in its preferred phase. A15 also scores each network as states and reports the remedy, lag-0 CMI given the phase.

	A11: idealised, events			A15: hidden co-drivers and noise, $\rho = 0.6$		
Locking λ	$\rho = 0.3$	$\rho = 0.6$	$\rho = 0.9$	Events	States	Remedy, events
0.5 (none)	0.518 [0.499, 0.539]	0.504 [0.484, 0.524]	0.467 [0.447, 0.486]	0.501 [0.480, 0.522]	0.496 [0.477, 0.516]	0.997
0.7	0.410 [0.392, 0.430]	0.251 [0.238, 0.265]	0.106 [0.099, 0.116]	0.255 [0.242, 0.268]	0.501 [0.481, 0.522]	0.997
0.85	0.236 [0.222, 0.252]	0.114 [0.106, 0.122]	0.026 [0.023, 0.029]	0.119 [0.111, 0.128]	0.500 [0.480, 0.520]	0.998
0.95	0.208 [0.195, 0.221]	0.099 [0.091, 0.108]	0.021 [0.018, 0.025]	–	–	–
1.0 (strict)	0.184 [0.172, 0.197]	0.088 [0.081, 0.096]	0.018 [0.015, 0.021]	0.118 [0.109, 0.129]	0.508 [0.488, 0.528]	0.999
Strict minus none	-0.334 [-0.358, -0.311]	-0.416 [-0.436, -0.395]	-0.450 [-0.470, -0.428]	-0.383 [-0.405, -0.360]		

A11, all fifteen conditions: remedy 0.991 to 1.000; positive control (lag-1 TE, delayed drivers) 0.997 to 1.000. A15, strict: states minus events $+0.390$ [$+0.368$, $+0.411$].

S.8.1.6 The neighbours of pass transistors

The pairs that lag 1 ranks most distinctively on the recording are the neighbours of pass transistors, the groups on the two sides of a pass transistor, a relation the reference omits (Table S28, Appendix S.8.1). Shared-parent pairs, which under the transistor-path definition are all pass neighbours (Section 4.4), have by far the largest mean lag-1 transfer entropy of any class (40.8 millibits against 8.4 for wired pairs), and the 82 clock-gated pass-neighbour pairs the largest of all, 62.0 millibits, with almost no contemporaneous dependence (3.5 millibits of lag-0 CMI; supplementary). Two-step paths, the indirect relations that pairwise measures are expected to confuse with wiring, are not enriched. The omitted relation explains the distinctive top of the lag-1 ranking but not its failure: counting all 228 pass-neighbour pairs as positives raises lag 1's group AUC only from 0.469 to 0.497 (Table S25).

S.8.1.7 Latch transfer in the NMOS circuits

In the random NMOS circuits of Section 4.6, whose latches are known by construction, lag-1 transfer entropy registers transfer from a latch's input to its storage node (0.858 on ten circuits, A8; 0.779 [0.615, 0.873] on 50 new circuits, A10), but the prediction that lag 1 would beat lag 0 on that relation is not met in either test (A10: -0.071 [-0.201, $+0.047$]), and from the storage node to the logic gate it feeds, a combinational relation, lag 0 exceeds lag 1 by 0.449 [0.343, 0.551] (Table S29, Appendix S.8.1). On these circuits, then, lag-1 transfer entropy registers latch transfer but is not specific to it, unlike on the chip; why the random circuits differ from the 6502 in this respect was not tested.

S.8.1.8 H3: direction in full

H3 is not supported, and the verdict is robust. Its primary statistic compares the orientation accuracy of the lag-1 asymmetry with that of the rule "the member with the lower event rate is the target". At the group level the asymmetry is worse than the rule at $k = 1$, 2 and 4 (0.623, 0.560 and 0.618 against 0.701) and indistinguishable from it at $k = 8$ and at the AIS-selected history (0.727 against 0.701, difference $+0.026$ [-0.015, $+0.070$]; Table S30). Orientation above chance, which H3 as worded does not claim, holds only through one hub: the clock group is the source of 265 of the 1,437 wired group pairs and the target of none, its event series is nearly constant, so transfer entropy into it is close to zero and the asymmetry orients its pairs correctly by construction; without them the group accuracies at $k = 1$, 2 and 4 fall to 0.538, 0.461 and 0.531 (Table S22). Against the predictors as pre-specified, the asymmetry would have read as supported, because the pre-specified rate predictor had the sign the wrong way round; reversing it after the run removed that apparent support. A logistic model fitted with the pre-specified weighting by physical edges finds that the asymmetry adds nothing to the nuisance

TABLE S34: Lag-0 CMI adds a small increment to the non-dynamical covariates at the group level, and the H2 contrast does not depend on them (supplementary; moved from Table 5.5), whole recording. Panel A: the scores of Table 5.5 with the clock group excluded as source and target (precision column: as source). Panel B: group AUC with comparisons only within deciles of a covariate, and the paired lag-0 minus lag-1 contrast at the group and physical levels. Panel C: increment of lag-0 CMI over a cross-validated logistic model of log group sizes, log distance, log event rates and phase class. Intervals are 95% target-cluster intervals; panels A and B are reproduced, and panel C recomputed, from the run's saved matrices (Appendix S.12).

A. Clock group excluded

Score	Group AUC	Group AP lift	Within-target AUC	Physical AUC	Precision at allocation (%)
Source-group size	0.834 [0.814, 0.854]	9.8 [7.8, 12.7]	0.782 [0.760, 0.802]	0.506 [0.478, 0.538]	10.3 [7.9, 13.4]
Layout distance	0.915 [0.901, 0.929]	12.9 [11.2, 15.0]	0.945 [0.936, 0.954]	0.860 [0.836, 0.884]	12.5 [10.6, 14.4]
Lag-0 CMI	0.682 [0.656, 0.708]	6.65 [5.17, 8.70]	0.761 [0.739, 0.781]	0.641 [0.598, 0.679]	13.9 [12.3, 15.4]

B. AUC within covariate strata

Strata	Lag-0 CMI, group	Lag-1 TE, group	Δ group	Δ physical
Source-group size	0.637 [0.615, 0.655]	0.431 [0.404, 0.459]	+0.206 [+0.175, +0.236]	+0.059 [+0.028, +0.094]
Layout distance between groups	0.606 [0.584, 0.628]	0.481 [0.456, 0.508]	+0.125 [+0.104, +0.147]	+0.055 [+0.025, +0.097]

C. Increment of lag-0 CMI over the covariate model

Model fitted at	Metric	Covariates alone	Increment from adding lag-0 CMI
Group level	Group AUC	0.969	+0.0041 [+0.0025, +0.0060]
Physical level	Physical AUC	0.890 [0.872, 0.908]	+0.005 [−0.002, +0.012]

predictors at any history; with group pairs weighted equally it adds a small increment that excludes zero at every history (Table S23), so the asymmetry carries a little orientation information that the rate rule does not, too little to beat it.

S.8.1.9 H4: signed influence in full

H4 is not supported, and as framed it is not testable on a change-coded recording. Its primary statistic, the difference in the percentage of negative signed influence δ between pull-down-only and unwired group pairs, is $+12.9$ points $[+8.3, +17.2]$ at the deciding group level, but $+3.3$ $[-9.1, +15.1]$ physically, so the levels disagree (Table S31). Within parity strata the group-level effect reverses: in every stratum pull-down-only pairs are *less* often negative than unwired pairs, and the composition-adjusted difference is -5.6 points $[-9.8, -1.7]$. The unstratified $+12.9$ is a composition effect, and the aliasing of Section 5.3.1 explains it: for a same-parity pair a source change at $t-1$ places t in the phase in which the target cannot change, so δ is negative whether or not the pair is wired, as it is for 91.8 per cent of unwired same-parity pairs, and pull-down-only pairs are 86 per cent same-parity against 42 per cent of all pairs. Nor does δ measure what H4 was framed to detect: under change coding an inverter's input and output change in the same sample, so a negative δ means only that a source change makes a target change in the next sample less likely.

S.8.1.10 Every written prediction of the account

Table S39 lists every written prediction of the account tested in Chapter 5, with its status and its outcome under the written decision rule; Section 5.10 summarises which held and which failed.

TABLE S35: Lag-1 transfer entropy registers the recording's latch pairs only at a lag of one sample, and orients them from the driven to the storage side (moved from Table 5.4). Hash-logged, exploratory tests (A5, A6, A13), change coding, whole recording. Panel A: stratified AUC of transfer entropy at source lag u against pairs with no wiring and no latch relation; A13 splits lags 2 to 7 by history coverage. Panel B: orientation by transfer-entropy asymmetry and by two nuisance rules, with paired intervals.

A. Lag profile (A5) and history coverage (A13)

Lag u	1	2	3	4	5	6	7	8
Latch pairs (41)	0.986	0.057	0.027	0.058	0.024	0.113	0.093	0.253
Pull-down wired pairs (802)	0.358	0.447	0.358	0.457	0.366	0.469	0.383	0.475

A13, lags 2 to 7, by history coverage: covered 0.038 [0.016, 0.077] (210 pair–lags); uncovered 0.239 [0.118, 0.376] (36); difference $+0.201$ [$+0.076$, $+0.334$]

B. Orientation (A6), driven to storage side

Rule	Correct (%)	TE asymmetry minus rule (points)
Lag-1 TE asymmetry	97.6 [92.7, 100]	–
Lower event rate is the target	58.5	$+39.0$ [$+31.7$, $+46.3$]
Higher conditional entropy is the target	47.6	$+50.0$ [$+36.6$, $+62.2$]

TABLE S36: Hash-logged tests on the simulated chip, change coding unless stated: stratified node-pair AUCs with 95% target-cluster intervals (1,000 draws) and orientation accuracies. Program 2 is the held-out program. Negatives for the latch rows are pairs with no measured influence in either direction. Latch pairs scored: 142 (program 1) and 170 (program 2). Longer lags (A13), delayed influence in general (A1, A16.2–3) and the wiring on program 2 (A7) are in Table S37.

Test	Program 1	Program 2	Prediction
Latch pairs under intervention (A1, A16)			
Latch pairs delayed-only, clamped with the latch closed (A16.1)	182 of 182	213 of 213	More than half: met
Latch pairs immediate, clamped with the latch open (supplementary)	182 of 182	213 of 213	–
Latch pairs in A1's delayed-only class, random moments	0 of 182	0 of 213	–
Latch pairs by transfer entropy (A1b, A5, A6)			
Lag-1 TE, driven to storage	0.978 [0.969, 0.987]	0.984 [0.974, 0.991]	Above 0.5: met, material
Lag-0 CMI, same pairs	0.586 [0.521, 0.640]	0.492 [0.437, 0.549]	Below lag 1: met
Lag-1 TE, storage to driven	0.220 [0.154, 0.293]	0.137 [0.095, 0.186]	Below 0.5: met
Node states: lag-1 TE; lag-0 CMI	0.985; 0.986	0.991; 0.991	–
Lag at which the AUC peaks	1 (0.978)	1 (0.984)	Lag 1: met
Orientation: TE; rate rule; entropy rule (%)	98.6; 52.1; 37.7	98.8; 55.0; 47.1	Above both rules: met

TABLE S37: On the simulated chip, latch pairs fall below chance at longer lags where the target's history covers the copy, delayed influence in general is registered only weakly, and lag-1 transfer entropy ranks the wiring below chance (moved from Table S36). Hash-logged tests, change coding: stratified node-pair AUCs with 95% target-cluster intervals (1,000 draws) and paired differences. Program 2 is the held-out program. A13: latch pairs at source lags 2 to 7, pooled over pair–lag combinations and stratified by lag; a target's history *covers* the copy at lag u if $k \geq u - 1$. ‡'Contradicted' for a paired difference extends the written rule, a choice made after the runs.

Test	Program 1	Program 2	Prediction
Lags 2 to 7, by history coverage (A13)			
Covered, latch pairs against no influence (A13.1)	0.228 [0.154, 0.310]	0.103 [0.067, 0.145]	Below 0.5: met
Uncovered (A13.3)	0.428 [0.301, 0.564]	0.474 [0.381, 0.571]	Upper limit ≥ 0.5: met
Uncovered minus covered (A13.2)	$+0.200$ [$+0.045$, $+0.361$]	$+0.371$ [$+0.267$, $+0.472$]	Above 0: met
Pair–lag combinations: covered; uncovered	797; 55	905; 115	
Delayed influence against none			
A1: lag-1 TE, random moments (primary)	0.571 [0.557, 0.585]	0.535 [0.523, 0.546]	Above 0.5: met, not material
A1: lag 1 minus lag 0	-0.016 [-0.024, -0.009]	-0.043 [-0.051, -0.034]	Above 0: contradicted‡
A16.2: lag-1 TE, timed moments	0.589 [0.571, 0.606]	0.537 [0.526, 0.547]	Above 0.5: met
A16.3: lag 1 minus lag 0	-0.021 [-0.027, -0.014]	-0.040 [-0.048, -0.032]	Above 0: contradicted‡
Wiring, wired against unwired node pairs (A7)			
Lag-1 TE	–	0.300 [0.276, 0.324]	At or below 0.5: met
Lag-0 CMI	–	0.575 [0.549, 0.603]	–

TABLE S38: A source block of two samples escapes the bound but recovers no wiring (moved from Table 5.2). Hash-logged test A18 on the recording, whole recording: AUCs of wired against unwired pairs with 95% target-cluster intervals (1,000 draws), for the block $(\Delta X_{t-1}, \Delta X_{t-2})$ and for lag-1 TE on the same samples, within strictly locked same-phase targets and over all group pairs. "Met" refers to the written decision rule; no paired test of block against lag 1 was fixed.

Statistic	Recording	Prediction
Block of two, within targets	0.460 [0.390, 0.533]	Lower limit ≤ 0.5: met
Lag-1 TE, same samples	0.400 [0.323, 0.480]	–
Block of two, all group pairs	0.472 [0.452, 0.492]	Lower limit ≤ 0.5: met
Lag-1 TE, same samples	0.469 [0.445, 0.491]	–

TABLE S39: Every written prediction of the account, with its outcome under the written decision rule. All are hash-logged. [e]Exploratory: the account was derived after the main results. [p]Partly anticipated by results already reported. [n]Near-certain given the design or the simulator, and so weak evidence. [h]Held out: no prediction was formed from the data. P1, P2: the simulated chip's programs; DK, PF, SI: Donkey Kong, Pitfall, Space Invaders. Values are AUCs or paired differences, with 95% target-cluster intervals where one value is given; precision differences are in percentage points.

Test	Prediction	Data	Result	Met
The mechanism (Section 5.3.2)				
A3[e,n]	Lag-1 TE never exceeds its bound	DK	0 of 77,006 strictly locked pairs	yes
A3[e]	At matched bound, wired and unwired rank alike (0.5)	DK	0.400 [0.373, 0.427]	no
A9.1[e]	Targets follow wired sources more	DK, P1, P2	0.725, 0.686, 0.719	yes ×3
A9.2[e]	Followership lowers wired sources below their activity	DK, P1, P2	-0.093, -0.082, -0.062	yes ×3
A9.3[e]	At matched bound, wired pairs go into targets with lower c	DK, P1, P2	0.454, 0.478 [0.448, 0.509], 0.351	yes, no, yes
A9.4[e]	Matching on c removes the matched-bound inversion	DK, P1, P2	$+0.028$, $+0.025$, $+0.066$	no, no, yes
A14.1[p]	Matching on q_0 and π raises the matched-bound AUC	DK, P1, P2	$+0.003$, -0.007, $+0.048$	no, no, yes
A14.2[p]	The matched AUC then includes 0.5	DK, P1, P2	0.454, 0.418, 0.428	no ×3
A12.1[p]	Mixed targets score above strict targets	DK	$+0.155$ [$+0.078$, $+0.223$]	yes
A12.2–3[p]	Strict below chance; mixed above	DK	0.449; 0.604	yes
A18.1–2[n]	A source block of two recovers no wiring (lower limits ≤ 0.5)	DK	0.472 (group); 0.460 [0.390, 0.533] (within targets)	yes
Held-out programs (Section 5.4)				
R2.1, 1b[h]	Lag-1 TE below chance (group); not above (physical)	PF, SI	0.397, 0.399; 0.438, 0.435	yes ×2
R2.2[h]	Lag 0 above lag 1	PF, SI	$+0.176$, $+0.188$	yes ×2
R2.3[h]	Most groups locked	PF, SI	85.8%, 85.7%	yes ×2
R2.4[h]	Followership above 0.5; penalty below 0	PF, SI	0.705, 0.709; -0.041, -0.043	yes ×2
R2.5[h]	Mixed above strict targets	PF, SI	$+0.147$, $+0.155$	yes ×2
R2.6, 6b[h]	Latch pairs found by lag 1, not lag 0	PF, SI	0.962, 0.967; 0.371, 0.329	yes ×2
R2.7[h]	Layout distance above lag-0 CMI	PF, SI	$+0.356$, $+0.343$	yes ×2
Beyond the chip (Section 5.5)				
A11.1–2	Strict locking ranks within-sample drivers below chance; none at chance	Synthetic	0.088 (strict); 0.504 (none) at $\rho = 0.6$	yes
A11	At chance without locking at every coupling	Synthetic	0.467 at $\rho = 0.9$	no
A15.1–2	Penalty with hidden co-drivers and noise; grows with locking	Synthetic	0.118; -0.383	yes
A15.3–4	State coding reduces the penalty; phase-conditioned lag 0 recovers drivers	Synthetic	$+0.390$; 0.997 to 0.999	yes
What lag 1 recovers (Section 5.6)				
A16.1[n]	Latch pairs delayed-only when clamped closed	P1, P2	182/182; 213/213	yes ×2
A20.1–3[e,n]	The same, in the recorded run	DK	41 of 41 pairs	yes
A16.2	Lag-1 TE ranks delayed-only influence above none	P1, P2	0.589; 0.537	yes ×2
A16.3	Lag 1 beats lag 0 on that class	P1, P2	-0.021; -0.040	no ×2
A13.1[p]	Covered latch pair–lags below chance	P1, P2	0.228, 0.103	yes ×2
A13.2[p]	Uncovered above covered	P1, P2, DK	$+0.200$, $+0.371$, $+0.201$	yes ×3
A13.3[p]	Uncovered not below chance	P1, P2, DK	0.428, 0.474, 0.239	yes, yes, no
A19.1	In-phase share ranks latch sources above others	DK	0.550 [0.498, 0.597]	no
A19.2	Latch transfer mostly in-phase; strictly locked pairs not	DK	0.946; 0.496	yes
Consequences for multivariate inference (Chapter 6)				
A4[e]	With the phase supplied: fewer same-phase sources; latch precision rises; netlist precision may not	DK	73% opposite-phase; latch unchanged; -0.7 [-1.9, $+0.3$]	yes; no; –
A17.1–2[n]	Lag-0 sources raise precision on the clock-aligned samples; equal to R1	DK	$+13.2$ [$+7.4$, $+19.2$]; $+0.07$ [-1.9, $+2.2$]	yes

TABLE S40: †Equal-size, per-target and allocation-matched comparators for the multivariate network, with target-cluster bootstrap intervals for the lift (Section 6.2).

Network	Level	Edges	Precision	Recall	Lift	CI low	CI high
IDTxl multivariate TE	group	1,443	0.039	0.039	7.260	5.412	9.183
IDTxl multivariate TE	physical	1,443	0.002	0.025	6.169	3.750	9.122
Lag-1 TE, global top-1,443	group	1,443	0.014	0.014	2.593	1.537	3.870
Lag-1 TE, global top-1,443	physical	1,443	5.20e-04	0.012	1.467	0.836	2.292
Lag-1 TE, per-target allocation matched to IDTxl	group	1,443	0.020	0.020	3.760	2.437	5.237
Lag-1 TE, per-target allocation matched to IDTxl	physical	1,443	0.001	0.016	3.655	1.922	6.088
Lag-1 TE, per-target top-1	group	519	0.048	0.017	9.012	5.615	12.6
Lag-1 TE, per-target top-1	physical	519	0.002	0.015	5.470	3.117	9.101
Lag-1 TE, per-target top-2	group	1,038	0.028	0.020	5.227	3.480	7.253
Lag-1 TE, per-target top-2	physical	1,038	0.001	0.016	3.597	2.180	5.318
Lag-1 TE, per-target top-3	group	1,557	0.020	0.022	3.725	2.498	5.062
Lag-1 TE, per-target top-3	physical	1,557	8.72e-04	0.017	2.460	1.439	3.903
Lag-1 TE, per-target top-5	group	2,595	0.013	0.024	2.451	1.691	3.310
Lag-1 TE, per-target top-5	physical	2,595	6.15e-04	0.017	1.737	1.080	2.677
Lag-0 CMI, global top-1,443	group	1,443	0.079	0.079	14.8	11.8	18.4
Lag-0 CMI, global top-1,443	physical	1,443	0.003	0.092	7.826	4.529	13.1
Lag-0 CMI, per-target allocation matched to IDTxl	group	1,443	0.139	0.140	26.1	22.6	29.3
Lag-0 CMI, per-target allocation matched to IDTxl	physical	1,443	0.005	0.130	14.8	9.297	24.4
Lag-0 CMI, per-target top-1	group	519	0.210	0.076	39.3	32.4	46.0
Lag-0 CMI, per-target top-1	physical	519	0.006	0.065	16.7	11.1	24.9
Lag-0 CMI, per-target top-2	group	1,038	0.152	0.110	28.5	24.1	33.1
Lag-0 CMI, per-target top-2	physical	1,038	0.005	0.106	13.2	9.031	19.1
Lag-0 CMI, per-target top-3	group	1,557	0.128	0.139	24.0	20.8	27.1
Lag-0 CMI, per-target top-3	physical	1,557	0.005	0.136	13.2	9.576	17.7
Lag-0 CMI, per-target top-5	group	2,595	0.101	0.183	19.0	16.6	21.2
Lag-0 CMI, per-target top-5	physical	2,595	0.003	0.165	8.337	5.997	11.9
Zero-lag MI, global top-1,443	group	1,443	0.080	0.080	14.9	11.8	18.5
Zero-lag MI, global top-1,443	physical	1,443	0.003	0.093	9.064	5.352	15.0
Zero-lag MI, per-target allocation matched to IDTxl	group	1,443	0.137	0.138	25.7	22.4	28.7
Zero-lag MI, per-target allocation matched to IDTxl	physical	1,443	0.005	0.140	15.0	9.679	25.4
Zero-lag MI, per-target top-1	group	519	0.214	0.077	40.0	33.2	46.5
Zero-lag MI, per-target top-1	physical	519	0.007	0.091	20.1	13.1	31.4
Zero-lag MI, per-target top-2	group	1,038	0.157	0.113	29.4	25.0	33.6
Zero-lag MI, per-target top-2	physical	1,038	0.005	0.120	15.4	10.9	22.2
Zero-lag MI, per-target top-3	group	1,557	0.130	0.141	24.3	21.0	27.3
Zero-lag MI, per-target top-3	physical	1,557	0.004	0.139	12.0	8.606	16.8
Zero-lag MI, per-target top-5	group	2,595	0.096	0.173	18.0	15.8	20.0
Zero-lag MI, per-target top-5	physical	2,595	0.003	0.163	8.651	6.579	11.7

S.9 Multivariate transfer entropy

TABLE S41: †Transistor-path relation classes of the group pairs selected by the H6 search (R1, the run of record: lags 0 and 1, 100 stratified targets; recomputed from its saved outputs), split by the lag at which each source was selected, and of pairwise lag-0 CMI allocated in the same way.

Network	Wired	Reverse only	Two-step	Shared parent	Shared child	Unrelated
IDTxl lags {0, 1}: lag-0 selections (178)	39	16	2	2	14	105
IDTxl lags {0, 1}: lag-1 selections (60)	2	0	0	10	0	48
IDTxl lags {0, 1}: all selections (238)	41	16	2	12	14	153
lag-0 CMI, allocation matched to all selections (238)	41	15	7	1	15	159

TABLE S42: The networks compared in Chapter 6, what each holds equal to the multivariate network, and what its contrast with that network isolates. All except the H6, A4, R1 and A17 networks are evaluated on the same 519 targets; edges are ordered group pairs. †Supplementary. ‡Hash-logged (Section 4.2).

Network	Edges	Held equal	Contrast isolates
IDTxl multivariate TE, source lag 1	1,443	–	–
IDTxl bivariate TE, link-wise FDR	122,441	Targets, embedding procedure, tests	Conditioning and size
Pairwise lag-1 TE, BH	89,625	Targets	Method and size
Pairwise lag-1 TE, global cut (H5 primary)	1,443	Targets, size	Method
Pairwise lag-1 TE, allocation-matched†	1,443	Edges into each target	Embedding and conditioning
Pairwise lag-1 TE at IDTxl's target lags, allocation-matched†	1,443	Edges into each target, target embedding	Conditioning
Lag-0 CMI, MI, rates, group size, distance†	1,443	Size; edges into each target if allocation-matched	Score
IDTxl, source lags 0 and 1, 100 stratified targets (H6 primary)	236	–	Lag admission
Lag-0 CMI, lag-1 TE, size and distance at the H6 allocation†	236	Edges into each target	Conditional search at lags 0 and 1
IDTxl, source lag 1, sample parity forced into the conditioning set, 100 targets (A4)‡	327	Targets, lag, embedding, tests	Supplying the phase
IDTxl, source lags 0 and 1, 100 re-drawn targets (R1)‡	238	The H6 design	The sample of targets
IDTxl, source lags 0 and 1, R1's targets, clock-aligned samples (A17)‡	237	The H6 design, R1's targets	The misaligned opening samples

TABLE S43: Paired increments in group precision (percentage points)† along the path of Figure 6.1a, with 95% target-cluster and two-way intervals and the share of draws in which the increment is positive. Values recomputed from the run's saved networks (Appendix S.12). The main run's own draws, which decide H5, give [+1.71, +3.39] for the total (Table 6.1); its vertex interval is in Table S50. The draws first reported gave [−0.06, +4.48] (97.3% positive) for conditioning: its lower limit lies at zero within resampling noise.

Contrast	Increment	Target bootstrap	Positive	Two-way bootstrap	Positive
IDTxl minus lag-1 global cut (total)	+2.49	[+1.62, +3.36]	100%	[+0.50, +5.05]	99.4%
Allocation: lag-1 allocation-matched minus global cut	+0.62	[+0.03, +1.21]	98%	[−0.35, +1.81]	90%
Embedding: at IDTxl's target lags minus at AIS histories	0.00	[0.00, 0.00]	–	[−0.06, +0.06]	–
Conditioning: IDTxl minus embedding-matched	+1.87	[+1.06, +2.66]	100%	[+0.09, +4.41]	98.2%
Lag-0 CMI, allocation-matched, minus IDTxl	+10.05	[+8.17, +11.92]	100%	[+6.47, +13.58]	100%

S.9.1 Supporting tables for Chapter 6

S.9.1.1 What the search did, in full

Three facts about the search bear on how its results are read (Table S46, Appendix S.9.1). First, for 334 targets (64 per cent) the largest selected target lag is 12, the cap, so the target's past was still informative at twelve phases and the cap may have truncated most embeddings. Second, parent-set size

TABLE S44: Relation classes declared by the multivariate network (IDTxl) and the allocation-matched lag-1 network (1,443 edges each)†, under both definitions of relation. Discordant: pairs declared by one network only. Ratio: multivariate enrichment over lag-1 enrichment, each relative to the network's rate for unrelated pairs, with target-cluster and two-way intervals. Exact: two-sided binomial test on the discordant pairs; Holm: corrected across the eight non-unrelated classes (reverse-only, two-step, shared-parent and shared-child under each definition; four have no discordant pair and an exact p of 1, which makes the correction conservative). Classes with no discordant pair are omitted. Of the multivariate network's 1,387 false positives, 1,320 are unrelated pairs and 63 shared-parent pairs; it has 27 fewer false positives than the lag-1 network (1,414), and under both definitions unrelated false positives fall by 37 net, but their discordant counts do not differ beyond chance.

Relation	Class size	Declared (IDTxl / lag 1)	Discordant	Ratio	Target interval	Two-way interval	Exact p	Holm p
Transistor paths								
Two-step	690	2 / 8	0 / 6	0.26	[0.09, 0.62]	[0.08, 1.13]	0.031	0.19
Shared parent	227	63 / 47	17 / 1	1.38	[1.19, 1.65]	[1.04, 1.97]	0.0001	0.001
Unrelated	263,896	1,320 / 1,357	783 / 820	–	–	–	0.37	–
Group graph								
Two-step	1,364	58 / 70	4 / 16	0.86	[0.73, 0.99]	[0.51, 1.25]	0.012	0.083
Shared parent	76,546	386 / 364	235 / 213	1.10	[1.02, 1.20]	[0.83, 1.47]	0.32	1.00
Unrelated	187,619	942 / 979	561 / 598	–	–	–	0.29	–

TABLE S45: The shared-parent relation and the multivariate network's 63 shared-parent false positives (group level)†, and group precision against the netlist with the 228 pass-neighbour pairs added, a reference defined after the errors were examined (95% target-cluster intervals). The 228 neighbour pairs include one that the transistor-path classes assign to the two-step class. The clamp test on the simulated chip (Section S.3.1.11) largely follows from the switch-level rule and never perturbed the recording's pairs, so it cannot show that each recorded pair is a transfer and not a proxy for a driver upstream of both sides.

Recording	Pairs	Of which clock-gated
Shared-parent group pairs	227	81
Multivariate shared-parent false positives	63	37
with lag-1 TE above lag-0 CMI	58	
whose parent's group is among the target's selected sources	8	
Pass-neighbour pairs, not wired	228	82
declared by the multivariate / allocation-matched lag-1 network	63 / 47	
declared by the lag-1 global cut / allocation-matched lag-0 network	27 / 4	
Netlist plus pass-through (post hoc)		Group precision (%)
Multivariate network		8.25 [6.77, 9.55]
Allocation-matched lag-1 network		5.27 [4.14, 6.38]
Allocation-matched lag-0 CMI network		14.21 [12.57, 15.74]

TABLE S46: IDTxl's target embeddings and parent sets by the target's AIS-selected history k, beside the circular-shift rejection rate of the pairwise test at $\alpha = 0.05$ with the target's past alone and with sample parity added (band means of the 40-target sample, frame-weighted for All; Table S16). Conditioning set: selected target lags plus selected sources. Threshold†: the analytic 95th percentile of the maximum over the remaining candidates that the last admitted source had to exceed (median over the band's targets). Wired lag-1 TE: mean lag-1 transfer entropy of wired pairs into the band's targets, physical-weighted. The number of selected target lags has a rank correlation of 0.90 with the AIS-selected k. Of the 519 targets, 515 received at least one source; the four without a source all have $k \leq 2$, and one of them is the clock group. [a]Over the 515 targets with at least one source. [b]All 54 targets with $k \leq 3$.

		Surrogate rejection (%)		Mean	Median	Mean	Median	Wired
k band	Targets	Past only	With parity	target lags	cond. set	sources[a]	threshold (bits)	lag-1 TE (bits)
1–3	54	37.0[b]	3.85[b]	1.35	4	3.08	0.00025	0.00007
4–6	130	38.9	1.9	3.44	6	2.84	0.00038	0.00081
7–9	78	34.3	0.59	6.45	9	2.71	0.0019	0.0034
10	77	21.4	0.006	8.84	12	2.84	0.012	0.0079
11–12	180	0.16	0	9.76	13	2.72	0.024	0.0164
All	519	20.3	0.89	6.67	10	2.80	0.0034	0.0084

barely varies with the target's history: across the history bands it lies between 2.71 and 3.08 sources while the median conditioning set rises from 4 to 13 binary variables, and it has a rank correlation of only 0.27 with the target's number of wired inputs. Third, with 200 surrogates no p-value can fall below 0.005, so every omnibus p-value sat at that floor and the network-level false-discovery control was vacuous as run; recomputed from the saved statistics with the analytic null (supplementary), every omnibus p-value is below 10^{-5} and the control keeps every target and edge, so for that stage the floor changed nothing, but the inclusion and pruning stages were not re-run.

What the stopping threshold means. Inclusion stops when the strongest remaining candidate falls below about the 95th percentile of the analytic null of the maximum. For 23 per cent of targets the last admitted source lies within a factor of 1.5 of that threshold, and only 1 of these 121 near-threshold admissions is wired. At $k = 11$ or 12, a third of the network, the threshold's median of 0.024 bits is 1.46 times the mean lag-1 transfer entropy of wired pairs into the same targets, so the threshold is comparable to the transfer a wired source carries, and the flat profile of parent-set sizes is not evidence that the search found the right number of parents. Exact significance does not identify which edges are wired: the 25 most significant selections include no wired pair.

TABLE S47: Networks of 1,443 group pairs on the same 519 targets†: group precision with 95% target-cluster interval, and physical precision (per cent). Allocation-matched networks give each target as many top-ranked sources as IDTxl selected for it. Whole recording unless marked clock-aligned. The paired increments along the path, with two-way intervals, are in Table S43 (Appendix S.9.1).

Network	Group precision	Phys. precision
IDTxl multivariate TE	3.88 [2.94, 4.83]	0.219
Lag-1 TE, global cut	1.39 [0.83, 2.07]	0.052
Lag-1 TE, allocation-matched	2.01 [1.34, 2.80]	0.130
Lag-1 TE at IDTxl's target lags, allocation-matched	2.01 [1.34, 2.80]	0.130
Lag-0 CMI, allocation-matched	13.93 [12.31, 15.47]	0.526
clock-aligned	14.15 [12.49, 15.72]	0.529
Lag-0 CMI given sample parity, allocation-matched	13.93 [12.32, 15.45]	0.497
Source-group size, allocation-matched	25.1 [22.7, 27.3]	–
Layout distance, allocation-matched	20.0 [17.7, 22.0]	–

TABLE S48: Secondary aims (H7, H8) and the descriptive test of mechanism 4, not decided. Panel A (H7): 80 whole-run lesions of randomly chosen simulated nodes, each node held low for 4,000 half-steps; a lesion is critical if the program's output differs from the reference (68 of 80). AUC of each node property in predicting criticality, without intervals; under no association an AUC from 68 critical and 12 intact lesions has a standard deviation of about 0.09. Panel B (H8): ratio of mean within-module to mean between-module score, the 97.5th percentile of a size-preserving label-permutation null (1,000 permutations; it preserves module sizes but not spatial clustering), and the agreement of Louvain communities (Blondel et al., 2008) with the module labels by normalised mutual information (NMI) and adjusted Rand index (ARI); recording at the group level, simulated chip on state-representation groups. Panel C: partial information decomposition of two-input NOR gates, medians across gates of the joint mutual information and of the fractions of it that are redundant and synergistic ($I_{\min}$); the recording's events are labelled rising edges in the run's outputs and are changes of state, as regeneration from source confirms (Section 4.3.1).

A. Lesions on the simulated chip: AUC predicting criticality

Node property	AUC
Mean outgoing lag-1 TE	0.55
Mean outgoing lag-0 CMI	0.55
Fraction of time high	0.48
Wired out-degree	0.51
Causal out-degree (same half-step)	0.77

B. Functional modules

Data	Network	Ratio	Null 97.5%	Communities	NMI	ARI
Recording	Lag-1 TE	1.09	1.08	5	0.14	0.08
	Zero-lag MI	2.14	1.09	6	0.26	0.18
	IDTxl multivariate	2.75	1.24	25	0.28	0.09
	Structural	13.02	1.22	11	0.54	0.40
Simulated chip	Lag-1 TE	1.08	1.05	7	0.11	0.04
	Lag-0 CMI	1.45	1.05	7	0.17	0.08
	Zero-lag MI	2.01	1.05	10	0.17	0.07
	Structural	8.45	1.21	13	0.48	0.25

C. Two-input NOR gates

Data	Representation	Lag	Gates	Joint MI (bits)	Redundant	Synergistic
Simulated chip	State	0	72	0.476	0.17	0.40
	State	1	72	0.137	0.14	0.38
	Rising edge	0	70	0.025	0.31	0.28
	Rising edge	1	70	0.025	0.31	0.29
Recording	As recorded	1	84	0.073	0.28	0.15

S.10 Secondary aims

TABLE S49: Decision register: one verdict per hypothesis, by the rule of Section 4.1.1. ‡Fixed after the final run. Robustness: two-way bootstrap and, where relevant, clock group excluded (Table S50). Values are group-level unless marked; intervals are 95% target-cluster. Clock-aligned: samples 1,408 onwards (Section 4.3.2).

	Primary statistic (null); deciding level	Verdict	Detail
H1	Lag-1 TE AUC (0.5) and AP lift (1); physical	Not supported (robust)	Physical AUC 0.50 [0.41, 0.60], AP lift 1.27 [0.99, 1.84]; clock-aligned AUC 0.44 [0.35, 0.53]. Group AUC 0.469 [0.445, 0.493], below chance under the target bootstrap only.
H2	Lag-0 CMI minus lag-1 TE in AUC (0); group‡	Partially supported (robust)	Comparative clause supported, and holds under the two-way bootstrap: $+0.163$ [$+0.142$, $+0.184$]; physical $+0.094$ [$+0.065$, $+0.128$]; unchanged on clock-aligned samples. Causal clause (settling within a step) not decidable from the recording (judgement).
H3	Orientation accuracy, lag-1 TE asymmetry minus the reversed event-rate rule‡, at $k = 1, 2, 4, 8$ (0); group	Not supported (robust)	Worse than the rate rule at $k = 1$, 2 and 4 under the target bootstrap; indistinguishable at $k = 8$. Not contradicted, because the condition does not hold at every history.
H4	Percentage of negative δ, pull-down-only minus unwired group pairs‡ (0); group	Not supported (robust)	Group $+12.9$ [$+8.3$, $+17.2$]; physical $+3.3$ [-9.1, $+15.1$], so the levels disagree. The effect reverses within every parity stratum. Not testable as framed on a change-coded recording.
H5	Group precision, IDTxl minus pairwise lag-1 TE cut at 1,443 group pairs (0); group‡	Not supported as worded (robust)	Precision gain present, and holds under the two-way bootstrap: $+2.49$ points [$+1.71$, $+3.39$] (main run; [$+1.62$, $+3.28$] in the re-drawn bootstrap of Table S50); physical agrees. Named mechanism absent (judgement). The pre-specified comparator is ill-posed: the lag-1 ranking does not order the wiring above chance.
H6	Group precision, IDTxl with source lags 0 and 1 minus lag 1, 100 stratified targets‡ (0); group	Supported (fragile: the physical interval includes zero under the two-way bootstrap)	$+13.13$ points [$+7.16$, $+19.23$] (4.1 to 17.2 per cent, R1, the run of record); physical $+0.47$ [$+0.10$, $+1.60$]. Same group precision as pairwise lag-0 CMI at the same allocation; below the source-size baseline there. Design fixed after the main run, and run after supplementary lag-0 analyses had been seen; an earlier draw, whose targets were not kept, gave $+13.28$.

S.11 Decision register

Table S49 gives one verdict per hypothesis by the rule of Section 4.1.1, and Table S50 gives the primary statistics under the three resampling schemes of Section 4.5. Both summarise results that Chapters 5 and 6 report in full, and they are placed here so that the design chapter states the rule without its outcomes.

TABLE S50: The primary statistics under three resampling schemes (95% percentile intervals, 1,000 draws each). Target-cluster: the deciding scheme. Two-way: sources and targets resampled independently (Owen, 2007). Vertex: one multiplicity vector over the 519 units applied in both roles (Snijders and Borgatti, 1999). Holm p: conservative bootstrap p-value, $\min\{1, 2(c+1)/(B+1)\}$ with c draws at or beyond the null, Holm-adjusted over H1 to H6 at the deciding level; 0.012 is the smallest value attainable with 1,000 draws, and H3 has one p-value for its four histories. H6 rests on R1, the run of record, recomputed from its saved outputs; its vertex interval was not recomputed (Appendix S.12). H5's target-cluster interval is the main run's, which decides it; its two-way and vertex intervals come from the supplementary draws. H6: 100 targets. Differences in precision and in percentage negative are in percentage points.

Hyp.	Statistic, level	Estimate	Target-cluster	Two-way	Vertex	Holm p
H1	Lag-1 TE AUC, physical	0.500	[0.411, 0.596]	[0.399, 0.617]	[0.418, 0.568]	1
	group (detail)	0.469	[0.445, 0.493]	[0.418, 0.528]	[0.422, 0.533]	–
H2	Δ AUC, group	$+0.163$	$[+0.142, +0.184]$	$[+0.102, +0.244]$	$[+0.099, +0.243]$	0.012
	physical	$+0.094$	$[+0.065, +0.128]$	$[+0.035, +0.214]$	$[+0.031, +0.233]$	–
H3	Δ accuracy, $k = 1$	-0.078	$[-0.120, -0.031]$	$[-0.174, +0.018]$	$[-0.173, +0.011]$	1
	$k = 2$	-0.141	$[-0.198, -0.076]$	$[-0.295, -0.002]$	$[-0.290, -0.008]$	
	$k = 4$	-0.083	$[-0.138, -0.026]$	$[-0.203, +0.034]$	$[-0.204, +0.024]$	
	$k = 8$	-0.003	$[-0.050, +0.047]$	$[-0.092, +0.093]$	$[-0.084, +0.087]$	
H4	Δ % negative, group	$+12.9$	$[+8.3, +17.2]$	$[+1.7, +21.1]$	$[+2.6, +21.1]$	0.012
H5	Δ precision, group	$+2.49$	$[+1.71, +3.39]$	$[+0.57, +5.20]$	$[+0.46, +5.35]$	0.012
	physical	$+0.17$	$[+0.10, +0.25]$	$[+0.06, +0.36]$	$[+0.05, +0.40]$	–
H6 (R1)	Δ precision, group	$+13.13$	$[+7.16, +19.23]$	$[+4.13, +22.05]$	–	0.012
	physical	$+0.47$	$[+0.10, +1.60]$	$[-0.23, +2.25]$	–	–

S.12 Recomputed supplementary results

The output files of the supplementary analyses (the activity-free baselines, the parity-conditioned scores, the decomposition of the multivariate network's precision and the enrichment of relation classes) and of the original H6 run were not kept. The supplementary values were recomputed from the main run's saved recording, score matrices and networks by a released script (analysis 09 of the code repository; Section 4.1.2), with the same target-cluster bootstrap (1,000 draws, seed 2026) and, for the two-way bootstrap, an independent multinomial draw of sources (seed 2027). A second pass recomputed the items whose first-pass definitions differed from those of the thesis (the tie handling of the phase-class ranking and the weighting of the physical-fitted covariate model); where the two passes differ, the second pass's values are used and the first pass's are superseded. Its self-test reproduces the main run's AUCs (0.469, 0.632 and 0.597 for lag-1 transfer entropy, lag-0 CMI and zero-lag MI). Every value the body uses is reproduced at the precision printed, or within 0.001, except those in Table S51. No difference changes a verdict.

A third pass recomputed the statistics behind the verdicts of H1 to H4 and their supplementary tables (Tables S50, S30, S22, S31 and the main rows of Table S24). With the main run's own bootstrap generator it reproduces every point value, every target-cluster interval and every Holm p-value at the precision printed. The two-way and vertex intervals were drawn afresh with the same schemes, because the code that drew the originals was not kept; their limits differ from those printed by at most 0.015 in AUC or accuracy and 0.5 points for H4, and no verdict or robustness label changes. The later supplementary scripts wrote some sixty output tables, and only seven survive, those written by the supplementary notebook. The values on which a conclusion of the body rests were recomputed. The rest are descriptive tables in this appendix and a few rows of the validation tables, and they report the values first computed; Table S1 marks each as *not retained*, and the scripts that produced them are not part of the evidence package.

The original H6 run cannot be recomputed, because its list of 100 targets was not saved. The H6 run of record is therefore R1, which drew the same stratified design with the same seed and allocation, recorded its targets, and was run under a hash-logged design. Its table (Table 6.3) was recomputed from R1's saved per-target outputs, which reproduce R1's logged values (238 sources, 41 wired, $+13.13$ points [$+7.16$, $+19.23$] over the main run's lag-1 network). The earlier draw gave 236 sources, 41 wired, 17.37 per cent.

S.13 Record of the final run

This record supports Section 4.1.2. Table S54, at the end of this section, classifies the thesis's headline claims by status (Section 4.1.2). Dates are file times (Table S52), versions and digests are in Table S53.

Timeline of the new tests (Section 4.2). The first wave's designs, statistics, decision rules and predictions were written in one file, saved at 22:11 on 2 November 2026. The runs began at 22:32 and ended at 02:01 the next morning. A second file, saved at 01:17 on 3 November, fixed the replication on a second program (A7), which ran from 02:02 to 02:17. Each script prints its file's SHA-256 hash at the start of its log, and both hashes match the files as they now stand. The A7 run's log prints the second file's hash; its output file records the first file's, because the chip script it calls begins by hashing the shared file. The files were not committed to version control or lodged with a registry, and the scripts were not hashed, so the hashes show that each file matches its run, not when the file was written.

Two further facts are disclosed. After the first file was saved and before the full runs, each script was checked on a small slice of the data, and every check wrote its statistics to a file: A1 on a tenth of the chip's data; A2, A3, A5 and A6 on the recording, including a latch-pair AUC of 0.99 on four positives; A8 on three short circuits; and R1 and A4 on two targets. No design, rule or prediction changed after those checks. A third file, saved at 14:33 on 3 November, fixed A9 and A10 after the other tests had run,

TABLE S51: †Supplementary values whose recomputation differs from the value first reported. All other recomputed values agree at the precision printed.

Quantity	First reported	Recomputed	Bearing
Covariate model, group-fitted: AUC; increment from lag-0 CMI	0.974; +0.0043 [+0.0028, +0.0059]	0.969; +0.0041 [+0.0025, +0.0060]	None: small positive increment
Covariate model, physical-fitted: AUC; increment	0.894; −0.0007 [−0.0049, +0.0033]	0.890; +0.005 [−0.002, +0.012]	None: interval includes zero
Rates model: group AUC; AP lift; within-target AUC	0.623; 10.39; 0.715	0.625; 8.75; 0.721	None: not used by a verdict
Conditioning increment, two-way interval (share positive)	[−0.06, +4.48] (97.3%)	[+0.09, +4.41] (98.2%)	Lower limit at zero within resampling noise: reported as marginal
Total increment, target-cluster and two-way intervals	[+1.62, +3.28]; [+0.57, +5.20] (99.8%)	[+1.62, +3.36]; [+0.50, +5.05] (99.4%)	None
Allocation increment, two-way interval	not reported	[−0.35, +1.81] (90%)	Allocation's share not established under resampling of sources

and both ran from 14:39 to 14:54. For these two the scripts were hashed as well: each log records the hashes of the design file, of the script and of the shared module. The only checks before the runs used synthetic data and a load check that printed array shapes, so no statistic of either test existed before its run. The file was not committed to version control either. A fourth file, saved at 15:21 on 3 November after A9's results, fixed A11, a test on synthetic networks, which ran from 15:22 to 15:37 under the same safeguards. A fifth file, `PREDICTIONS_A12.md`, saved on the night of 3 November, fixed A12 to A19; it states that several predictions were partly informed by descriptive results already reported, and which. The runs started at 23:59 on 3 November and finished at 01:56 on 4 November, A17 taking the two hours; every log records the hashes of the design file, the script and the shared module, and the checks before the runs used synthetic data or printed only shapes and timings. The supplementary open-latch check for A16 was run after its results had been seen; it was rerun on 4 November to report the share of delayed-only pairs in each clamp set. A sixth file, `PREDICTIONS_R2.md`, saved at 13:53 on 4 November after the recordings had been regenerated and before any statistic was computed on them, fixed the replication; both runs started at 14:15 and finished by 14:24, and each log records the hashes of the design file and the script. A seventh, `PREDICTIONS_A20.md`, saved at 14:40, fixed the timed clamps on the recorded run. The first attempt stopped when the simulator failed to settle under one clamp; an addendum saved at 14:42, before any statistic had been computed, fixed how such clamps are recorded (as non-convergent, with the simulator rebuilt and restored), and the run went from 14:42 to 14:47 with both hashes logged. The roles of the author and of the software tools in drafting these files are stated in the Acknowledgements.

Health report. The final computation is one execution of the notebook from a clean results directory with seed 2026. A health report, required to pass before any result was reported, checks twelve conditions, and all passed; among them:

- the thesis configuration was used, not the smoke-test configuration;
- the vectorised estimator agreed with JIDT to within 10^{-9} bits (observed 3.2×10^{-16});
- the structural reference had 3,543 evaluable edges;
- the simulated processor executed its program to completion, with every output byte correct (10,460 instructions);

- the 160 synthetic Boolean and 80 NMOS tasks completed;
- IDTxl was installed, ran without failures and was complete for all 519 targets under both methods;
- every required figure and table was written.

Interruptions and changes. The run was interrupted twice and resumed from cached stage outputs, so three sessions contribute. The executed notebook differs from the version tested in the smoke runs in two cells. In cell 80 of 107 (index 79) the IDTxl lag-sensitivity stage was given two workers instead of six after six had stalled in 16 GB of memory; the worker count should not affect results, although this was not tested. In cell 85 (index 84) the simulator's check of the shift-register output was changed to replay the program's writes with wraparound, because the register loop wraps after 256 writes and the original check rejected a correct run. The replay matched all 256 bytes, a deliberately flipped bit makes it fail, and the change was approved before the final relaunch and alters no number.

Reused files. The per-target files of mutual information, storage and lag-1 transfer entropy were written during the first smoke session and reused. The cells that compute them have no quick-mode branch, and the files were re-verified: the vectorised estimator reproduced the lag-1 matrix to 3.2×10^{-16} bits, and an independent export recomputed five targets' transfer entropies, five pairs' mutual information and every event rate to within 3.1×10^{-16} bits.

Provenance. The notebook writes the toolkit and simulator modules it imports, so the code executed is the code recorded, and the toolkit's digest is the same in the evidence packages of 29 and 30 October. Every table is written as CSV and LaTeX, every figure as PDF and PNG, and a results ledger records each quoted number with its configuration. The manifest records the digests of the recording and the JIDT jar. IDTxl was installed from its public repository at commit `c7eacfd`, and all 55 installed files match the hashes in its installation record, so every change to its behaviour is made at run time. IDTxl's own settings record gives its version as unknown, so version and commit are taken from the installation record and the ledger; it needs two packages it does not declare, mpmath and pyopencl. The executed notebook is that of the final resumed session and shows cached stages, so timings come from per-target records and file times. An evidence package with the executed notebook, code, configuration, health report and all outputs accompanies the thesis.

Departures from the proposal. The research proposal (INFO4990) planned to extract half-cycle node states from JSSim, the JavaScript simulator behind Visual6502, and to analyse subsystem-restricted candidate pools. The project instead adopted the event recording of Jonas and Kording (2017), analysed every active unit and takes node states from its own simulator. A graded circuit-semantic ground truth was replaced by sharply defined references whose disagreements are measured, and search-aware surrogate testing by a fixed history-selection rule and delay set.

Environments and cost. The final run used Python 3.10 as x86-64 code under Rosetta emulation, for compatibility with the Java bridge. IDTxl dominates the cost: 59,022 seconds summed over targets for the multivariate inference (median 117 seconds per target; 2 hours 44 minutes of wall time on six workers) and 44,634 seconds for the bivariate inference (about 2 hours 5 minutes). Lag-1 transfer entropy for all pairs took 616 seconds summed over targets in JIDT and 28 seconds with the vectorised estimator. The supplementary analyses ran on a separate two-core Linux machine with the same JIDT jar and IDTxl commit, except five computed by a supplementary notebook in the main run's environment. That machine also ran the stratified H6 run on two workers (median 247 seconds per target, no failures), which reproduces the main run's selections exactly for a check target.

TABLE S52: Dated record of the analysis before and during the final run (Section 4.1.2). The project has no version-control commits, so every date is a file time on the project machine. "801 units" is the earlier grouping of the unreduced recording; "reduced" is the recording of Section 4.3.1.

Date	Evidence
30 Jul 2016	File date of the recording
2–3 May 2026	Research-proposal slide decks (contents not inspected for hypothesis wording)
6 Aug 2026	Request to the recording's authors for documentation (no reply found)
21 Sep 2026	Netlist file `transdefs.js` added
24 Sep 2026	Version control initialised (no commit ever made); first mutual-information, transfer-entropy and ROC pilots scored against the wiring; a pilot of thresholds
25 Sep – 7 Oct	Histories up to $k = 20$; fixed $k = 15$ and 18; pooled storage; full pairwise transfer entropy with Benjamini–Hochberg control
8 – 14 Oct	Surrogate pilots (up to 999 surrogates); runs on the 801 units, including IDTxl multivariate inference
14 – 26 Oct	Pairwise runs on the reduced recording, method versions 2 to 4, three with history allowances labelled 1.25, 1.10 and 1.05; earlier notebook created 15 Oct
26 Oct	Evidence packages of the earlier work; earlier notebook last edited (no pre-specified hypotheses in either)
29 Oct, 15:27	Final notebook created: first dated appearance of H1–H8
29 Oct, 15:30 – 16:16	Run manifest; three smoke runs in quick mode; `segdefs.js` downloaded; core matrices written
29 Oct, 16:16 – 21:34	Final run starts; synthetic benchmarks and the IDTxl multivariate, bivariate and sensitivity runs complete
30 Oct, 01:17 – 01:20	Lag-sensitivity stage stalls on memory; workers reduced to two; relaunch
30 Oct, 02:49	Run stops at the simulator's shift-register check
30 Oct, 08:01	Check changed to replay writes with wraparound; relaunch
30 Oct, 09:35	Run complete; health report passes 12 of 12 checks
31 Oct – 1 Nov	Supplementary analyses from saved outputs; stratified H6 run (targets drawn 1 Nov, 00:30; complete 1 Nov, 04:06)

TABLE S53: Computing environment, software versions and file digests of the final run (Section 4.1.2 and Appendix S.13, p. xlviii). Digests are SHA-256, abbreviated to their first and last characters.

Item	Value
Machine	MacBook Pro, Apple M1 Pro (8 cores: 6 performance, 2 efficiency), 16 GB memory
Operating system	macOS 26.2 (Darwin 25.2.0)
Python	3.10.20 (conda-forge), x86-64 under Rosetta emulation
Numerical packages	NumPy 1.26.4, SciPy 1.11.4, mpmath 1.4.1, pandas 2.1.4, scikit-learn 1.7.2, statsmodels 0.14.6, NetworkX 3.4.2
Java bridge and runtime	JPype1 1.7.1; Zulu OpenJDK 17.0.18, x86-64
JIDT	1.6.1, `infodynamics.jar`, digest `2d367c24...5b11`; identical to the jar bundled with IDTxl
IDTxl	1.6.1, commit `c7eacfd`; the 55 installed files match the installation record; changes made at run time only
Undeclared IDTxl dependencies	mpmath; pyopencl (installed only so that the estimator lookup does not exit; OpenCL not used)
Toolkit module	Written by the notebook; digest `5433c744...`, identical on 29 and 30 October
Recording	`donkeykong.5000.ts.spikes.npy`, 2,246,396,570 bytes, 639,999 × 3,510 `uint8`, digest `ae34d1fe...fcfec`
Netlist	`transdefs.js` (file date 21 Sep 2026); `nodenames.js` (16 Oct 2026); `segdefs.js` (downloaded 29 Oct 2026 from the main branch, no commit pinned)
Supplementary analyses and the stratified H6 run	Separate Linux machine, x86-64, 2 cores: Python 3.13.18, NumPy 2.5.3, SciPy 1.18.1, pandas 3.0.5, mpmath 1.4.1, JPype1 1.7.1, OpenJDK 21.0.12; the same JIDT jar (digest `2d367c24...`) and IDTxl commit `c7eacfd`; five analyses (S1–S5) from a supplementary notebook in the main environment
Random seed	2026 (bootstrap, calibration targets and shifts); 7 for the source draws of the two-way bootstrap
Version control	None during the run; the code, design files and outputs were later placed in a repository whose provenance folder reproduces every logged hash (Section 4.1.2)

TABLE S54: Register of the thesis's headline claims by status. Pre-specified: hypothesis and primary test fixed before the final run. Exploratory: computed by the final run, testing no hypothesis. Supplementary: computed afterwards from the run's saved outputs, with its results in view. Exploratory also covers a result resting on a reference or rationale formed after the main results. Hash-logged: a new test whose design, statistic, decision rule and prediction were written in a file before its run, with the run's log recording the file's SHA-256 hash (Section 4.2). Verdicts are in Table S49.

Claim	Status	Section
RQ1: pairwise inference		
Lag-1 TE does not rank wiring above chance at the physical level (H1)	Pre-specified	5.2
Lag-1 TE ranks wiring below chance at the group level and within targets under the target-cluster bootstrap	Exploratory; within-target, two-way and vertex intervals supplementary	5.2
Ranking by significance does no better than ranking by TE	Exploratory; exact log tails supplementary	5.2
The lag-1 failure does not come from the selected histories	Supplementary	5.3
Aliasing by clock phase accounts for the failure in same-parity pairs	Supplementary	5.3.1
Lag 1 ranks pass-transistor neighbours, which the reference omits	Supplementary	S.8.1.6
The unreduced sweep's apparent recovery of wiring is an artefact of the row offset, and the recording carries no usable timing below one phase	Exploratory sweep; row-offset control supplementary	S.8.1.2
Lag-0 CMI identifies wiring better than lag-1 TE (H2, comparative clause)	Pre-specified; prediction possibly informed by zero-lag MI computed since 24 September; metric and level chosen after	5.7
Most of lag 0's AUC is phase membership; its own contribution lies at the top of the ranking	Supplementary	5.7
The H2 contrast survives controls for parity, rates, the clock group, common histories and source resampling	Supplementary	S.8.1.4
The first 1,407 reduced samples are out of step with the clock, which inflated the contemporaneous scores; on clock-aligned samples lag 0 orders only each target's top few sources	Supplementary	4.3.2
Source-group size and layout distance rank the wiring better than every information measure; lag 0 adds a small increment to them	Supplementary	5.9
TE asymmetry does not orient wired pairs beyond the event-rate rule (H3)	Pre-specified; comparator and null chosen after	5.8
TE asymmetry orients wired pairs above chance, through the clock group's pairs	Descriptive	5.8
Pull-down edges are not shown to be suppressive; the group-level effect is a composition effect of parity (H4)	Pre-specified; parity strata supplementary	5.8
Contemporaneous scores recover mostly pull-down edges	Supplementary	5.9
RQ2: multivariate inference		
IDTxl is more precise than the lag-1 ranking cut at its size (the precision gain of H5)	Pre-specified statistic	6.2
Allocation and conditioning, not embedding, account for the gain	Supplementary	6.2.1
Conditioning does not robustly remove indirect-path errors, and shared-parent errors rise (the mechanism of H5)	Pre-specified clause without statistic; evidence supplementary	6.4
Lag-0 CMI at IDTxl's allocation is several times as precise as IDTxl	Supplementary	6.2
Network-level false-discovery control is vacuous at 200 surrogates	Exploratory	6.3
Admitting lag-0 sources improves multivariate recovery (H6)	Pre-specified hypothesis; stratified design chosen after the main run	6.6
With lag 0 admitted, IDTxl is no more precise than pairwise lag-0 CMI given the same allocation	Supplementary	6.6
Data, estimators and replication		
One sample is one clock phase, and an event marks a change of state	Supplementary; confirmed at source by regeneration (Section 4.3.1)	4.3.1
The surrogate rejection rate of the analytic test is high at short histories, mostly from phase dependence, and the test is conservative at long histories	Exploratory; parity and placebo re-analysis supplementary	4.6
Replications on the simulated chip and NMOS circuits, including the clamp test of pass transistors	Exploratory; change coding, intervals, activity baseline and the clamp test supplementary	4.6
New tests fixed in writing before they were run (Section 4.2)		
The bound of Proposition 2 holds pair by pair; wired same-phase pairs rank below unwired ones at matched bound (A3)	Hash-logged, exploratory; prediction of 0.5 not met	5.3.2
Within a target, lag-1 TE ranks strictly locked same-phase sources exactly by their change rate when the target is unchanged, and followership lowers wired sources (A9; recording and both chip programs)	Hash-logged, exploratory	5.3.2

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